UNIVERSITY SCHOOL OF AUTOMATION & ROBOTICS (USAR)

SMART INDIA HACKATHON 2026 - MASTER PROBLEM STATEMENTS CATALOG

Official Administrative Evaluation Document | Generated: Sunday, 23 August 2026 | Total PS: 66

1. [GGSIPU2601] AI-Powered Enterprise Multi-Agent Workflow Orchestration Platform

Category: Software	**Theme:** Agentic AI & Enterprise Automation
Organization: University School of Automation & Robotics (USAR), GGSIPU	**Domain:** Multi-Agent Systems, LLMs, Enterprise AI, Workflow Automation

Problem Statement:

Modern enterprises operate across multiple disconnected software platforms, resulting in fragmented workflows, manual coordination, and operational inefficiencies. Develop an enterprise-grade multi-agent platform capable of autonomously planning, executing, monitoring, and optimizing business workflows. The platform should support specialized AI agents collaborating through shared memory and reasoning while securely integrating with enterprise applications via APIs or Model Context Protocol (MCP). It should include workflow orchestration, human-in-the-loop approvals, explainable AI, audit logging, and cloud-native deployment.

Expected Solution:

A scalable multi-agent AI platform supporting autonomous task planning, secure enterprise integrations, workflow execution, monitoring dashboards, approval mechanisms, memory management, and explainable decision-making.

Technologies & Dataset:

Tech: Gemini, LangGraph, CrewAI, AutoGen, Python, Go, PostgreSQL, Redis, Neo4j, Pinecone, Docker, Kubernetes, Google Cloud | **Dataset:** Public APIs, Enterprise Sample Data, Synthetic Datasets

Deliverables:

Source Code, Architecture Diagram, Deployment Guide, API Documentation, Demo Video, Technical Report

2. [GGSIPU2602] Immersive 3D VR/AR Digital Marketplace for Handloom Industry and Workers in India

Category: Software	**Theme:** Digital Economy & Smart Industry
Organization: University School of Automation & Robotics (USAR), GGSIPU	**Domain:** AR/VR, 3D, E-Commerce, Handloom, AI

Problem Statement:

Develop an immersive digital marketplace enabling Indian handloom weavers, artisans, cooperatives, buyers and retailers to showcase, discover, customize and trade handloom products through interactive 3D, AR and VR experiences. The platform should digitally represent products, artisans and weaving processes while improving the digital reach of traditional handloom workers.

Expected Solution:

Build a scalable 3D/AR/VR marketplace with virtual showrooms, interactive 3D product visualization, 360° views, AR-based product placement and VR exhibitions. Include artisan onboarding, digital cataloguing, product metadata, AI-powered recommendations and visual search, multilingual assistance, buyer-seller communication, order management, analytics and optional virtual handloom experiences. The platform should support smartphones, browsers and VR devices and remain usable in low-bandwidth environments.

Technologies & Dataset:

Tech: Unity, Unreal Engine, WebXR, Three.js, Babylon.js, ARCore, ARKit, WebGL, Blender, Python, FastAPI, PostgreSQL/MongoDB, Computer Vision, Recommendation Systems, Generative AI, LLM/RAG, NLP | **Dataset:** Handloom product images/videos, 3D product models, fabric textures, artisan metadata, weaving techniques, GI-tagged product information, product catalogues and user interaction data. Teams may create an annotated 3D handloom product dataset.

Deliverables:

Functional web/mobile marketplace, 3D product catalogue, AR visualization module, VR showroom, artisan portal, AI recommendation/search system, multilingual assistant, buyer-seller interaction module, analytics dashboard, 3D asset library, technical documentation, usability study and demonstration video.

3. [GGSIPU2603] Development of Surface Air Quality Index (AQI) and Identification of Formaldehyde (HCHO) Hotspots over India Using Satellite Data

Category: Software	**Theme:** Environment, Climate & Sustainability
Organization: University School of Automation & Robotics (USAR), GGSIPU	**Domain:** Remote Sensing, Geospatial AI, Deep Learning, Earth Observation, Atmospheric Science

Problem Statement:

Air pollution remains one of the leading environmental and public health challenges in India, causing significant health risks and economic losses. Most regions lack sufficient ground-based monitoring stations, making it difficult to assess real-time air quality across the country. Satellite observations from INSAT-3D, Sentinel-5P (TROPOMI), and other Earth Observation missions provide continuous atmospheric measurements that can be utilized to estimate ground-level pollutant concentrations. Develop an AI-driven geospatial platform capable of estimating Surface Air Quality Index (AQI) from satellite-derived Aerosol Optical Depth (AOD), meteorological parameters, and ground observations. Additionally, the solution should identify and analyze Formaldehyde (HCHO) hotspots associated with biomass burning, forest fires, and agricultural residue burning by integrating multi-source satellite imagery, fire count datasets, and atmospheric transport models. The platform should generate high-resolution spatial maps, identify pollution hotspots, analyze pollutant transport, and support policymakers in air quality management under the National Clean Air Programme (NCAP).

Expected Solution:

Develop a complete AI-enabled geospatial analytics platform capable of integrating satellite observations, CPCB monitoring data, meteorological datasets, and fire activity information. The solution should estimate surface pollutant concentrations using deep learning models (CNN, LSTM, CNN-LSTM or Transformer-based models), compute AQI across India, identify HCHO hotspots through spatial clustering and statistical analysis, correlate pollution with biomass burning events, analyze pollutant transport using wind fields, and generate interactive dashboards with daily high-resolution air quality and hotspot maps.

Technologies & Dataset:

Tech: Python, Google Earth Engine, TensorFlow, PyTorch, CNN, LSTM, CNN-LSTM, Vision Transformers, GIS Libraries (GDAL, Rasterio, GeoPandas), QGIS, ArcGIS, OpenCV, Plotly, Streamlit | **Dataset:** INSAT-3D Aerosol Optical Depth (MOSDAC), Sentinel-5P (TROPOMI NO_2, SO_2, CO, O_3 & HCHO), CPCB Ground Air Quality Data, MODIS/VIIRS Active Fire Data, ERA5, IMDAA, MERRA-2 Meteorological Data

Deliverables:

AI model for Surface AQI estimation, High-resolution AQI maps over India, HCHO hotspot detection system, Biomass burning impact assessment, Pollution source region identification, Interactive GIS Dashboard, Technical Report, Source Code, Model Documentation

4. [GGSIPU2604] Agentic AI Platform for Intelligent Customer Acquisition, Digital Adoption & Personalized Banking Engagement

Category: Software	**Theme:** Digital Banking & Financial Inclusion
Organization: University School of Automation & Robotics (USAR), GGSIPU	**Domain:** Agentic AI, Generative AI, Large Language Models (LLMs), Machine Learning, Conversational AI, Recommendation Systems, NLP, Customer Analytics, Intelligent Automation

Problem Statement:

Banks today face increasing challenges in acquiring customers efficiently, improving adoption of digital banking products, and maintaining long-term customer engagement. Traditional banking journeys often involve fragmented onboarding, low product discovery, generic customer communication, and limited personalization, leading to reduced customer satisfaction and lower digital adoption rates. Develop an Agentic AI-powered intelligent banking platform capable of autonomously assisting customers throughout their banking lifecycle—from customer acquisition and KYC onboarding to personalized financial guidance, digital product recommendations, proactive engagement, and intelligent financial assistance. The platform should leverage autonomous AI agents that collaborate to understand customer intent, analyze behavioral patterns, recommend suitable financial products, automate customer interactions, and deliver contextual, secure, and personalized banking experiences while ensuring compliance with banking regulations and responsible AI practices.

Expected Solution:

Develop an enterprise-grade multi-agent AI platform that autonomously manages customer onboarding, verifies KYC documents, recommends suitable banking products, predicts customer intent, provides conversational financial assistance, generates personalized offers, monitors customer engagement, and proactively guides users toward increased adoption of digital banking services. The solution should integrate customer analytics, recommendation engines, workflow automation, multilingual conversational AI, fraud-aware interactions, and human-in-the-loop escalation mechanisms to deliver secure and personalized banking experiences.

Technologies & Dataset:

Tech: Python, LangGraph, CrewAI, AutoGen, Gemini, OpenAI, Llama, TensorFlow, PyTorch, FastAPI, PostgreSQL, MongoDB, Redis, Vector Databases (Pinecone/ChromaDB), Docker, Kubernetes, Google Cloud/AWS/Azure | **Dataset:** Synthetic Banking Data, Customer Transaction Data (Anonymized), Product Catalogs, CRM Data, Customer Journey Logs, Public Financial Datasets, RBI Guidelines, Open Banking APIs (where applicable)

Deliverables:

Functional AI Banking Assistant, Multi-Agent Workflow Engine, Personalized Recommendation System, Customer Engagement Dashboard, Banking Analytics Portal, API Documentation, Technical Report, Source Code, Deployment Guide, Demonstration Video

5. [GGSIPU2605] AI-Powered Inclusive Learning & Wellbeing Platform for Neurodivergent Learners

Category: Software	**Theme:** Inclusive Education, Accessibility & Digital Health
Organization: University School of Automation & Robotics (USAR), GGSIPU	**Domain:** Artificial Intelligence (AI), Generative AI, Agentic AI, Machine Learning, Natural Language Processing (NLP), Computer Vision, Speech AI, Human-Computer Interaction (HCI), Accessibility Technologies, Affective Computing, Personalized Learning

Problem Statement:

Neurodivergent individuals, including those with ADHD, Autism Spectrum Disorder (ASD), Dyslexia, Dyspraxia and Sensory Processing Disorders, often encounter educational technologies designed for neurotypical learning patterns, resulting in reduced engagement, learning barriers, communication challenges and limited accessibility. In an engineering school such as USAR, GGSIPU East Delhi Campus, this shows up concretely: dense lecture slides, timed lab assessments, group-project coordination, viva and presentation anxiety, noisy laboratories, and examination formats that penalise processing-speed differences rather than conceptual understanding. Develop an AI-powered inclusive platform that adapts educational content, communication, emotional support or creative workflows to the unique cognitive and sensory needs of neurodivergent learners in a higher-education setting. The solution should prioritize accessibility from the design stage and leverage AI to deliver personalized learning experiences, assistive communication, emotional wellbeing support or creative expression. Teams must demonstrate a user-centered design approach, ideally co-designed with the university's Equal Opportunity Cell / student counselling services and with neurodivergent students themselves, under informed consent.

Expected Solution:

Develop an intelligent AI platform capable of dynamically personalizing learning materials, assisting communication, supporting emotional wellbeing, or enhancing creative expression based on the user's cognitive profile and preferences. The solution may include adaptive content generation, AI tutors, emotion-aware assistants, multimodal interaction (voice, text, gestures, images), accessibility customization, intelligent task management, or social skills coaching. The system should continuously learn from user interactions while maintaining privacy, explainability, and inclusive user experience principles.

Technologies & Dataset:

Tech: Python, TensorFlow, PyTorch, OpenAI/Gemini/Llama, LangGraph, CrewAI, Whisper, TTS/STT APIs, OpenCV, MediaPipe, Flutter/React, FastAPI, Firebase, PostgreSQL, Vector Databases, Google Cloud/AWS/Azure | **Dataset:** Public Educational Datasets, Learning Analytics Data, Synthetic Neurodiversity Scenarios, Emotion Recognition Datasets, Speech & Accessibility Datasets, User Feedback Data (with consent), Open Educational Resources

Deliverables:

Functional AI Application, Accessible User Interface, Personalized Recommendation Engine, Technical Documentation, Source Code, Demo Video, User Testing Report, Architecture Diagram, Deployment Guide

6. [GGSIPU2606] Computer Vision-Based Smart Examination Monitoring System

Category: Software	**Theme:** Smart Education & Examination Integrity
Organization: Examination Division, GGSIPU	**Domain:** Computer Vision, Edge AI, Artificial Intelligence, Deep Learning, Video Analytics, Biometrics, Human Activity Recognition

Problem Statement:

Ensuring the integrity of mid-semester examination and end-semester examination for regular unbiversity student and Comman entrance test conducted by Examination Division is a recurring operational challenge for GGSIPU where a single examination shift can involve thousands of candidates across multiple halls with a limited pool of invigilators drawn from teaching duty. Impersonation, unauthorized electronic devices, inter-candidate communication, seat-plan violations and suspicious movement are difficult to detect reliably through manual observation alone, and any subsequent unfair-means (UFM) case must be defensible before an examination committee. Develop a privacy-preserving AI-based examination monitoring system capable of continuously analysing CCTV or edge-camera feeds installed in university examination halls to detect malpractice in real time. The system must generate evidence in a form that supports the university's existing UFM reporting procedure, minimise storage of personally identifiable information, and comply with applicable privacy regulations and university policy.

Expected Solution:

Develop an AI-powered invigilation platform capable of real-time candidate monitoring, suspicious activity detection, liveness verification, unauthorized object detection, automated alert generation, examination analytics, and post-exam incident reporting. The system should leverage edge computing for low-latency inference, minimize cloud dependency, support multiple camera feeds, and provide explainable evidence for every detected violation.

Technologies & Dataset:

Tech: Python, OpenCV, YOLOv11/YOLOv12, MediaPipe, TensorFlow, PyTorch, ONNX Runtime, Edge TPU, NVIDIA Jetson, FastAPI, PostgreSQL, Docker | **Dataset:** AI City Challenge, COCO Dataset, Open Images, Custom Examination Surveillance Dataset, Face Recognition Datasets, Synthetic Video Data

Deliverables:

Smart AI Invigilation Platform, Violation Detection Dashboard, Incident Reports, Edge Deployment Model, Source Code, Technical Documentation, Demo Video

7. [GGSIPU2607] AI-Based Campus Safety Intelligence System

Category: Software	**Theme:** Smart Campus & Public Safety
Organization: General Administration Branch	**Domain:** Computer Vision, Edge AI, Artificial Intelligence, Deep Learning, Video Analytics, Biometrics, Human Activity Recognition

Problem Statement:

The GGSIPU East Delhi Campus accommodates thousands of students, faculty members, staff, contractual workers and daily visitors, with peak loads during counselling, fests, examinations and placement drives. Existing CCTV coverage depends on manual observation by a small security team, so incidents such as scuffles, crowd crush at gates and canteens, unattended baggage, laboratory fire or smoke, unauthorized entry into restricted labs and workshops, and late-night intrusion are often noticed only after the fact. Develop an AI-powered campus safety intelligence platform capable of continuously monitoring campus surveillance feeds to detect accidents, violence, overcrowding, unattended baggage, fire, unauthorized access, intrusion into restricted areas and other safety threats in real time. The platform should intelligently prioritise incidents, generate automated alerts routed to the campus security control room and the designated safety officer, support emergency response, and preserve privacy by processing video at the edge wherever possible.

Expected Solution:

Build a scalable AI surveillance platform capable of real-time incident detection, crowd density estimation, intrusion detection, fire and smoke recognition, violence detection, abandoned object identification, emergency alert generation, and centralized campus safety dashboards. The solution should integrate edge AI devices, GIS-based campus visualization, and event logging to improve situational awareness and emergency response.

Technologies & Dataset:

Tech: Python, OpenCV, YOLOv11/YOLOv12, TensorFlow, PyTorch, DeepSORT, MediaPipe, NVIDIA Jetson, Edge TPU, MQTT, FastAPI, PostgreSQL, Docker | **Dataset:** AI City Challenge Dataset, CrowdHuman, UCF Crime Dataset, Fire & Smoke Datasets, COCO, Custom Campus CCTV Data, Synthetic Surveillance Videos

Deliverables:

Campus Safety Dashboard, AI Surveillance Engine, Incident Alert System, GIS Visualization Portal, Edge Deployment Model, Technical Report, Source Code, Demo Video

8. [GGSIPU2608] Low-Cost Lite Smart Hostel Allocation & Optimization System using Google Apps Script

Category: Software	**Theme:** Smart Campus Management & Student Services
Organization: GGSIPU Hostels	**Domain:** Artificial Intelligence (AI), Optimization Algorithms, Google Apps Script, Workflow Automation, Decision Support Systems, Data Analytics

Problem Statement:

Hostel allocation at GGSIPU Dwarka Campus and East Delhi Campus is a complex annual exercise involving students from multiple programmes with varying eligibility criteria, room preferences, accessibility requirements, gender policies, reservation norms under university rules, roommate compatibility, academic year, branch, distance from hometown and special accommodation requests. The process is currently driven by spreadsheets and physical counters, resulting in inefficient room utilization, allocation conflicts, lengthy grievance processes and limited transparency for students and parents. Develop a low-cost, lightweight Hostel Allocation & Optimization System using Google Workspace technologies that automates the complete hostel admission lifecycle — from online applications and document verification to optimized room allocation, waiting-list management, room upgrades, hostel transfers, vacancy tracking and grievance redressal. The solution must be operable by a small university administrative team without dedicated servers, paid licences or a full-time developer.

Expected Solution:

Develop a web-based Hostel Allocation & Management System using Google Apps Script and Google Sheets as the primary backend. The platform should support online hostel applications through Google Forms, automated eligibility verification, preference-based room allocation, waiting-list optimization, roommate compatibility analysis, hostel transfers, vacancy management, grievance handling, and administrator approvals. The allocation engine should use Google OR-Tools or custom optimization algorithms to generate fair and policy-compliant allocations while Google Sheets serves as the centralized database. The system should automatically generate allotment letters, send notifications through Gmail, maintain audit logs, provide administrator dashboards, and offer students a self-service portal for application tracking and hostel-related requests. The complete solution should be deployable within the Google Workspace ecosystem using only free or open-source technologies, making it suitable for educational institutions with limited IT infrastructure and budget.

Technologies & Dataset:

Tech: Google Apps Script, Google Sheets, Google Forms, Google Drive, Gmail API, Google Charts, HTML, CSS, JavaScript, Bootstrap/Tailwind CSS, Google OR-Tools (or Custom Optimization Algorithms), Python (Optional), FastAPI (Optional), SQLite/PostgreSQL (Optional), GitHub | **Dataset:** Student Application Data, Hostel Room Inventory, Student Preference Forms, Historical Allocation Records, Institutional Hostel Policies, Google Forms Responses, Synthetic Test Data

Deliverables:

Smart Hostel Management Portal, Automated Allocation Engine, Student Self-Service Portal, Administrator Dashboard, Waiting List & Vacancy Management Module, Automated Email Notification System, Technical Documentation, Source Code Repository, Deployment Guide, Demo Video

9. [GGSIPU2609] Low-Cost Lite Blockchain-Based Certificate Generation, Verification & Management System using Google Apps Script

Category: Software	**Theme:** Digital Governance, Smart Campus & Document Security
Organization: Directorate of Students' Welfare, Various Dep	**Domain:** Blockchain, Cryptography, Google Apps Script, Web Technologies, Automation, Information Security, Digital Identity

Problem Statement:

The Directorate of Students' Welfare and Guru Gobind Singh Indraprastha University and various University Schools (USS)/ Centre issues thousands of certificates every academic year hackathon and Smart India Hackathon internal-round certificates, workshop and FDP certificates, internship and industrial-training certificates, society and event participation certificates, merit and achievement certificates, and bona fide/character certificates — in addition to the university's degree certificates and transcripts. These are currently produced through manual mail-merge, signed physically, and verified over email or telephone whenever an employer or another institution asks, making the process slow, inconsistent and vulnerable to forgery. Commercial certificate platforms are expensive and require dedicated infrastructure, which most university schools cannot sustain. Develop a low-cost, lightweight certificate lifecycle management platform using Google Apps Script and Google Workspace that automates certificate creation, approval by the competent authority, digital signing, blockchain-based hashing, secure storage, QR-code verification, revocation management and public validation deployable directly on the university's existing Google Workspace domain.

Expected Solution:

Develop a web-based Certificate Management System using Google Apps Script and Google Sheets as the primary backend. The platform should support certificate template creation, bulk certificate generation from Google Sheets, automatic PDF generation, QR code embedding, unique certificate ID generation, SHA-256 blockchain-compatible hash creation, secure storage in Google Drive, email distribution via Gmail, and a public verification portal. Every issued certificate should generate a cryptographic hash stored in an immutable verification ledger to detect tampering. The system should also support certificate revocation, audit logs, role-based access control, approval workflows, analytics dashboards, and API integration with university portals, while remaining fully deployable using free or open-source technologies without dedicated servers.

Technologies & Dataset:

Tech: Google Apps Script, Google Sheets, Google Drive, Google Docs, Google Slides (Templates), Gmail API, HTML, CSS, JavaScript, Bootstrap/Tailwind CSS, QR Code Generator Libraries, SHA-256 Cryptographic Hashing, Blockchain Ledger (Optional: Polygon, Ethereum Testnet or Hyperledger Fabric), GitHub, Firebase (Optional), SQLite/PostgreSQL (Optional) | **Dataset:** Student Records, Certificate Templates, Event Registration Data, University ERP Data, Google Forms Responses, Historical Certificate Records, Synthetic Test Data

Deliverables:

Certificate Management Portal, Bulk Certificate Generator, QR-Based Verification Portal, Blockchain Hash Verification Module, Administrator Dashboard, Public Verification Portal, Audit Log System, Technical Documentation, Source Code Repository, Deployment Guide, Demo Video

10. [GGSIPU2610] Lightweight Academic Minor & Major Project Allocation and Evaluation Management System

Category: Software	**Theme:** Education & Academic Administration
Organization: University School of Automation & Robotics / GGSIPU	**Domain:** Web Application, Workflow Automation, Database Management, Education Technology

Problem Statement:

Develop a lightweight centralized system to manage the complete lifecycle of Minor Projects, Major Projects and 8th-semester Internships. For Minor Projects, students may work individually or in teams of a maximum of two students from the same batch/group (B1 or B2). The system should manage academic sessions, faculty capacity, project preferences, faculty-allotted students, project titles/domains, internal evaluations and marks. Students should be able to submit faculty preferences along with their proposed project title/domain, while faculty members should be able to directly pre-select students who have already discussed projects with them. The system must automatically maintain faculty-wise slot capacity and prevent over-allocation. For Major Projects/Internships, students should be able to select either one Major Project or one Internship in the 8th semester. Major Project allocation should follow the defined faculty/team rules, while students opting for internships should be mapped to an assigned faculty coordinator/supervisor. Faculty-in-charge and authorized administrators should be able to configure evaluation stages, panels, marks, deadlines, batches and academic sessions from a centralized interface.

Expected Solution:

Develop a Google Workspace-based lightweight portal using Google Apps Script and Google Sheets as the backend. Provide separate student, faculty, panel evaluator and administrator interfaces. Implement student registration and preference submission, faculty pre-allotment, automatic slot/capacity management, preference-based allocation, project title/domain collection, Major Project/Internship selection, faculty supervisor assignment, team formation, academic-session management, configurable evaluation stages, panel allocation, marks entry, moderation/finalization and automated reports. The system should provide role-based access, audit logs, notifications, dashboards and downloadable allocation/marks reports without requiring paid software licenses or dedicated server infrastructure.

Technologies & Dataset:

Tech: Google Apps Script, Google Sheets, Google Forms/HTML Service, JavaScript, HTML/CSS, Google Drive, Gmail/MailApp, Google Workspace, Apps Script Triggers | **Dataset:** Student master data, batch/group information, faculty list, faculty capacity/slots, academic sessions, project preferences, project titles/domains, faculty pre-allotments, project teams, internship records, evaluation stages, panel details and marks.

Deliverables:

Student portal, faculty portal, administrator dashboard, faculty slot/capacity management, preference and allocation module, pre-allotment module, Minor/Major/Internship workflow, team management, project title/domain repository, evaluation and panel module, marks entry system, automated reports, notifications, audit trail, Google Sheets database structure, deployment documentation and user guide.

11. [GGSIPU2611] AI-Assisted Automated Internal Examination Marks Entry and Verification System

Category: Software	**Theme:** Smart Education & Process Automation
Organization: Examination Division, GGSIPU	**Domain:** Agentic AI, RPA, Web Automation, Data Processing, OCR

Problem Statement:

Develop an intelligent browser-based agent or extension to automate the entry of internal examination marks into the university examination portal. Faculty members currently need to manually enter marks for every student, which is time-consuming and prone to errors. The system should accept marks from structured Excel/CSV files, identify and match students using enrolment number and name, locate the corresponding student record in the examination portal and populate the appropriate marks field automatically. The system must handle variations in student ordering, missing records, duplicate names, formatting differences and absent/invalid marks while ensuring that incorrect marks are never submitted automatically.

Expected Solution:

Develop a secure browser extension or agent that allows an authorized faculty member to upload an Excel/CSV marks file, automatically parse and validate the records, match enrolment numbers and names with students displayed on the examination portal, highlight unmatched or ambiguous records, populate the corresponding marks fields and provide a verification preview before final submission. The system should support dry-run mode, error reporting, configurable mark validation rules, progress tracking, automatic logging and rollback/recovery mechanisms. Actual submission should require explicit faculty confirmation.

Technologies & Dataset:

Tech: JavaScript/TypeScript, Chrome/Edge Extension APIs, Python, Playwright/Selenium, HTML/CSS, ExcelJS/SheetJS, OCR, fuzzy matching, NLP, RPA, Agentic AI, Google Sheets/Excel integration | **Dataset:** Sample anonymized student enrolment numbers, names, subject/course codes and internal marks in Excel/CSV format; sample portal page structures or a controlled mock examination portal; intentionally mismatched, duplicate and missing records for validation.

Deliverables:

Browser extension/AI agent, Excel/CSV import module, student matching engine, portal field-mapping and automation module, validation engine, preview/confirmation interface, unmatched-record report, audit logs, error recovery mechanism, mock portal for testing, technical documentation, security/privacy guidelines and demonstration video.

12. [GGSIPU2612] Smart Urban Sustainability & Resource Optimization Platform for Large-Scale Public Events

Category: Software	**Theme:** Smart Cities, Sustainability & Urban Planning
Organization: University School of Automation & Robotics (USAR), GGSIPU	**Domain:** Artificial Intelligence (AI), Machine Learning, Geospatial Analytics (GIS), Optimization, Predictive Analytics, Digital Twin, Data Science, Data Visualization

Problem Statement:

Large-scale public events such as Kumbh Mela, religious yatras, international sporting events, concerts, festivals, exhibitions, marathons, rallies and summits place significant pressure on urban infrastructure, transportation, utilities, emergency services, healthcare, waste management and environmental resources, yet city administrators rarely have an integrated decision-support platform to plan for them. The same problem occurs in miniature, and can be studied safely, on a university campus: admission counselling days, convocation, Anugoonj each bring several thousand additional people onto GGSIPU, straining parking, gates, canteens, washrooms, power load, waste generation and medical readiness. Develop an AI-powered Urban Sustainability & Resource Optimization Platform that integrates geospatial, transportation, environmental, IoT and demographic datasets to support intelligent planning and operational management of large-scale public events, and demonstrate it on a real university-scale gathering as a controlled testbed before generalising to city scale.

Expected Solution:

Develop a web-based decision support platform capable of integrating transportation, weather, GIS, satellite imagery, IoT, demographic, environmental, and event datasets. The solution should leverage AI and optimization algorithms to forecast crowd movement, estimate resource demand, optimize transportation routes, detect congestion hotspots, simulate planning scenarios, identify sustainability risks, monitor urban heat islands, and provide emergency response recommendations. The platform should include interactive GIS maps, dashboards, heatmaps, analytics, and sustainability scorecards to assist administrators before, during, and after large-scale public events.

Technologies & Dataset:

Tech: Python, Google Earth Engine, Google Maps Platform, Google Apps Script, Google Sheets, Looker Studio, Streamlit, Plotly, Leaflet.js, OpenStreetMap, GeoPandas, Scikit-learn, TensorFlow, FastAPI, PostgreSQL/PostGIS, Docker, GitHub | **Dataset:** OpenStreetMap, Census Data, Weather APIs, Satellite Imagery, Government Open Data, Traffic & Mobility Data, Air Quality Data, Energy & Water Consumption Data, Event Attendance Data, Public Transport Data, Synthetic Smart City Datasets

Deliverables:

Smart Urban Planning Dashboard, GIS Visualization Portal, Crowd Mobility Prediction Module, Transportation Optimization Engine, Sustainability Analytics Dashboard, Resource Consumption Forecasting Module, Scenario Simulation Tool, Technical Documentation, Source Code Repository, Deployment Guide, Demo Video

13. [GGSIPU2613] AI-Powered Intelligent YouTube Creator Automation & Analytics Platform

Category: Software	**Theme:** Creator Economy, Digital Media & AI Automation
Organization: University School of Automation & Robotics (USAR), GGSIPU	**Domain:** Artificial Intelligence (AI), Generative AI, Agentic AI, Computer Vision, Natural Language Processing (NLP), Data Analytics, Recommendation Systems, Automation

Problem Statement:

Content creators spend a significant amount of time on repetitive and time-consuming tasks such as content planning, script generation, SEO optimization, thumbnail creation, caption generation, video scheduling, audience engagement, comment moderation, analytics interpretation, and channel performance optimization. These manual workflows reduce productivity and limit creators' ability to focus on producing high-quality content. Develop an AI-powered YouTube Creator Automation Platform that streamlines and automates multiple stages of the YouTube content lifecycle. The platform should intelligently assist creators with idea generation, SEO optimization, automatic metadata creation, captioning, thumbnail recommendations, upload scheduling, audience analytics, comment moderation, and performance insights. The solution should leverage AI agents to automate repetitive workflows while providing actionable recommendations that improve content discoverability, audience engagement, and channel growth.

Expected Solution:

Develop a web-based AI platform capable of automating end-to-end YouTube creator workflows. The solution should support AI-assisted content ideation, script summarization, title and description generation, SEO keyword recommendations, thumbnail quality analysis, automatic caption generation, multilingual translation, upload scheduling, comment sentiment analysis, spam detection, analytics dashboards, and intelligent recommendations for improving audience retention and channel performance. The platform may integrate with the YouTube Data API while complying with API policies and rate limits and should demonstrate measurable improvements in creator productivity through workflow automation.

Technologies & Dataset:

Tech: Python, Google Apps Script (Optional Dashboard), Google Sheets, YouTube Data API v3, YouTube Analytics API, Gemini/OpenAI/Llama, TensorFlow, OpenCV, Whisper, FFmpeg, FastAPI, Streamlit, HTML, CSS, JavaScript, Bootstrap/Tailwind CSS, PostgreSQL/SQLite, GitHub | **Dataset:** YouTube Data API, YouTube Analytics API, Public YouTube Metadata, Video Caption Datasets, Open Thumbnail Datasets, Public Comment Datasets, Synthetic Creator Workflow Data

Deliverables:

AI Creator Dashboard, Workflow Automation Engine, SEO Optimization Module, Thumbnail Analysis Tool, Caption Generator, Comment Moderation System, Analytics Dashboard, Technical Documentation, Source Code Repository, Deployment Guide, Demo Video

14. [GGSIPU2614] Instagram Reel & YouTube Shorts Understanding and Semantic Search Engine

Category: Software	**Theme:** AI for Digital Content Discovery
Organization: University School of Automation & Robotics (USAR), GGSIPU	**Domain:** Vision-Language Models (VLM), Computer Vision, Natural Language Processing (NLP), Retrieval-Augmented Generation (RAG), Generative AI, Multimodal AI

Problem Statement:

Current short-video platforms such as Instagram Reels and YouTube Shorts primarily rely on captions, hashtags, keywords, and creator-provided metadata for content discovery. This limits users' ability to search videos based on their actual visual scenes, spoken dialogue, objects, activities, emotions, or contextual meaning. Develop an AI-powered multimodal semantic search engine capable of understanding every aspect of a short-form video, including video frames, speech, OCR text, background music, detected objects, human activities, emotions, scene context, and captions. The platform should enable users to perform natural language searches such as "Show reels where someone repairs a motorcycle while explaining the process" or "Find cooking videos where the chef uses an air fryer without mentioning it in the caption" without relying solely on hashtags or metadata.

Expected Solution:

Develop an intelligent semantic search platform that extracts multimodal embeddings from videos using Vision-Language Models, Automatic Speech Recognition, OCR, audio understanding, and object detection. The platform should index video content into a vector database and support conversational semantic search using RAG and LLMs. It should identify actions, objects, people, emotions, landmarks, products, spoken content, and contextual relationships while providing highly relevant search results, explainable retrieval, multilingual support, and similarity-based recommendations.

Technologies & Dataset:

Tech: Python, FastAPI, OpenCV, YOLOv11/YOLOv12, Whisper, BLIP-2, CLIP, LLaVA, Florence-2, Gemini API, Llama, LangChain, FAISS/Milvus/Pinecone, PostgreSQL, ElasticSearch, OCR (EasyOCR/Tesseract), FFmpeg, Docker, GitHub | **Dataset:** Public YouTube Videos, Instagram Public Content (where permitted), MSR-VTT, ActivityNet, YouCook2, COCO, LAION, Open Images Dataset, Synthetic Multimedia Dataset

Deliverables:

Semantic Video Search Engine, Multimodal Embedding Pipeline, AI Search Assistant, Vector Database, Video Understanding Dashboard, Technical Documentation, Source Code Repository, Deployment Guide, Demo Video

15. [GGSIPU2615] AI-Based Virality Prediction & Content Optimization Engine for Instagram Reels and YouTube Shorts

Category: Software	**Theme:** AI for Creator Economy & Social Media Analytics
Organization: University School of Automation & Robotics (USAR), GGSIPU	**Domain:** Computer Vision, Natural Language Processing (NLP), Time Series AI, Generative AI, Predictive Analytics, Multimodal AI

Problem Statement:

Content creators currently rely on intuition and historical analytics to estimate how a short-form video will perform after publishing. There is no intelligent system capable of accurately predicting engagement before content goes live. Develop an AI-powered multimodal virality prediction platform capable of estimating a video's watch time, audience retention curve, engagement rate, likes, comments, shares, saves, click-through rate, and overall virality probability before publication. The platform should analyze video quality, editing style, pacing, scene transitions, facial expressions, speech, captions, hashtags, trending audio, thumbnail quality, posting schedule, audience preferences, and historical engagement to provide actionable recommendations that improve discoverability and maximize audience reach across platforms such as Instagram Reels and YouTube Shorts.

Expected Solution:

Develop a multimodal AI platform that combines Computer Vision, NLP, Audio Analysis, Time Series forecasting, and Generative AI to predict content performance before publishing. The solution should analyze visual aesthetics, editing rhythm, spoken content, sentiment, topic relevance, hashtags, music trends, audience behaviour, and historical analytics to generate virality scores and optimization recommendations. The platform should provide explainable AI insights, predictive dashboards, A/B content comparison, posting time recommendations, trend forecasting, and AI-generated suggestions for improving titles, captions, hashtags, thumbnails, and overall engagement potential.

Technologies & Dataset:

Tech: Python, FastAPI, OpenCV, YOLOv11/YOLOv12, Whisper, TensorFlow, PyTorch, XGBoost, LSTM, Transformer Models, Gemini API, Llama, LangChain, Prophet, Scikit-learn, Plotly, Streamlit, PostgreSQL, YouTube Analytics API, FFmpeg, Docker, GitHub | **Dataset:** Public YouTube Analytics Data, Trending Video Datasets, Social Media Engagement Datasets, Open Video Caption Datasets, Public Creator Analytics, Synthetic Engagement Dataset

Deliverables:

Virality Prediction Engine, Creator Analytics Dashboard, AI Content Optimization Assistant, Trend Forecasting Module, Explainable AI Reports, Technical Documentation, Source Code Repository, Deployment Guide, Demo Video

16. [GGSIPU2616] Lightweight AI-Powered Intelligent Government Notesheet Generation & Decision Support System using Institutional Knowledge with Google Apps Script

Category: Software	**Theme:** e-Governance, Intelligent Document Automation & Decision Support
Organization: Office of the Dean, USAR	**Domain:** Agentic AI, Generative AI, Natural Language Processing (NLP), Retrieval-Augmented Generation (RAG), Knowledge Graphs, Document Intelligence, OCR, Workflow Automation

Problem Statement:

Government departments and public universities generate thousands of official note sheets, sanction proposals, administrative approvals, expenditure statements, office orders and financial justifications every year. University School of Automation & Robotics is direct instance of this: purchase of laboratory equipment, hiring of guest faculty, sanction of fest and hackathon expenditure, conference travel and TA/DA, project consumables under funded research, and creation of new posts each require a note sheet drafted with knowledge of previous cases, university statutes and ordinances, General Financial Rules (GFR), delegation of financial powers, purchase procedures, audit observations and administrative precedent. The work is time-consuming, inconsistent, and dependent on a handful of experienced officials, so files stall whenever those officials are unavailable. Develop an AI-powered Intelligent Notesheet Generation & Decision Support System that learns from historical note sheets, office memoranda, sanction orders, previous approvals, audit observations, circulars and institutional records to automatically draft context-aware note sheets on new matters, retrieve similar precedent cases, recommend wording, generate expenditure estimates and budget tables, reference applicable rules, suggest the correct approval chain up to the competent authority, identify missing documents and provide explainable justifications in official drafting style. The proposed system must be lightweight, cost-effective, and easy to deploy and maintain within an educational or government institutional environment. The solution should preferably be developed using Google Apps Script and existing Google Workspace services avoiding the requirement for dedicated servers, complex infrastructure, expensive software, paid enterprise licenses, or specialized hardware. The system should maximize the use of existing institutional resources and provide a practical, scalable, and user-friendly solution while keeping human officials in the decision-making and final approval loop.

Expected Solution:

Develop a secure AI-powered document intelligence platform that ingests historical note sheets, office orders, financial sanctions, circulars, acts, regulations, and institutional documents. Using RAG, Knowledge Graphs, and Agentic AI, the platform should identify similar precedent cases, retrieve applicable government rules (such as GFR, DFPR, university statutes, purchase procedures, audit guidelines, and office policies), automatically draft note sheets in official government format, generate expenditure statements and budget calculations, recommend competent approving authorities, detect missing approvals or supporting documents, and assist officers in preparing complete administrative files. The system should support multilingual drafting, approval workflow integration, explainable references, version control, document comparison, and continuous learning from newly approved cases while ensuring data security and confidentiality.

Technologies & Dataset:

Tech: Python, FastAPI, LangChain, LangGraph, LlamaIndex, Gemini API/OpenAI/Llama, RAG, FAISS/Milvus/Pinecone, Neo4j Knowledge Graph, OCR (Tesseract/EasyOCR), Google Apps Script (Optional), Google Workspace APIs, PostgreSQL, Elasticsearch, Docker, GitHub, PDF Processing Libraries | **Dataset:** Historical Government Note Sheets, Office Orders, Circulars, GFR & Financial Rules, University Statutes, Audit Reports, Budget Files, Purchase Records, Administrative Orders, Previous Approval Files, Public Government Documents, Synthetic Administrative Dataset

Deliverables:

Intelligent Notesheet Generator, Similar Case Retrieval Engine, Government Knowledge Base, Budget & Expenditure Estimation Module, AI Decision Support Dashboard, Rule Recommendation Engine, Approval Workflow Assistant, Technical Documentation, Source Code Repository, Deployment Guide, Demo Video

17. [GGSIPU2617] Agentic AI-Powered Multimodal Women's Health Intelligence & Personalized Care Platform

Category: Software	**Theme:** Digital Healthcare, Preventive Care & Personalized Medicine
Organization: University School of Automation & Robotics (USAR), GGSIPU	**Domain:** Agentic AI, Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Computer Vision, Natural Language Processing (NLP), Predictive Analytics, Digital Twin, Healthcare AI, Multimodal AI

Problem Statement:

Women's healthcare involves multiple interconnected stages throughout life, including menstrual health, PCOS, endometriosis, fertility, pregnancy, postpartum recovery, menopause, hormonal disorders, nutrition, mental wellbeing, and chronic disease management. Existing healthcare applications are often fragmented, focusing on isolated functions such as symptom tracking or appointment scheduling without providing continuous, personalized, evidence-based healthcare support. They rarely integrate multimodal medical information, wearable health data, laboratory reports, medical records, imaging, lifestyle habits, and conversational interactions into a unified decision-support ecosystem. Develop an Agentic AI-powered Multimodal Women's Health Intelligence Platform capable of creating a personalized digital health profile for every individual through continuous analysis of symptoms, laboratory investigations, prescriptions, wearable devices, nutrition, sleep, activity, hormonal cycles, stress levels, and clinical history. The platform should proactively identify potential health risks, provide explainable recommendations, support preventive healthcare, assist clinicians in decision-making, and improve long-term health outcomes while ensuring privacy, transparency, and responsible AI practices.

Expected Solution:

Develop an intelligent healthcare platform consisting of multiple autonomous AI agents collaborating to perform specialized healthcare tasks, including Symptom Assessment, Clinical Knowledge Retrieval, Laboratory Report Interpretation, Medical Document Intelligence, Nutrition Planning, Mental Wellness Support, Medication & Adherence Monitoring, Risk Prediction, Personalized Lifestyle Coaching, Appointment Management, Emergency Escalation, and Follow-up Care. The solution should leverage Retrieval-Augmented Generation (RAG) using trusted medical guidelines, research literature, and national healthcare protocols to provide evidence-based recommendations. It should maintain a personalized Digital Health Twin, integrate wearable devices and health sensors, interpret laboratory reports and medical documents using Document AI, utilize Computer Vision where applicable for medical image or visual health assessment, predict future health risks using AI models, generate personalized care plans, support multilingual voice and text interactions, provide clinician dashboards, maintain longitudinal patient records, and continuously improve recommendations while preserving data privacy and regulatory compliance.

Technologies & Dataset:

Tech: Python, FastAPI, LangGraph, LangChain, LlamaIndex, Gemini API/OpenAI/Llama, TensorFlow, PyTorch, OpenCV, YOLOv11/YOLOv12, Whisper, OCR (EasyOCR/Tesseract), FAISS/Milvus/Pinecone, Neo4j Knowledge Graph, PostgreSQL, FHIR/HL7 Standards, Google Fit/Health Connect APIs, Docker, GitHub | **Dataset:** MIMIC-IV (where applicable), PhysioNet, Women's Health Datasets (PCOS, Menopause, Pregnancy), WHO & MoHFW Clinical Guidelines, Nutrition Databases, Sleep & Activity Data, Laboratory Report Samples, Electronic Health Records (Synthetic), Medical Imaging Datasets (where applicable), Public Healthcare Knowledge Bases

Deliverables:

AI Healthcare Assistant, Multi-Agent Orchestration Engine, Personalized Digital Health Twin, Medical Knowledge Retrieval System, Laboratory Report Analyzer, Clinical Decision Support Dashboard, Risk Prediction Module, Doctor & Caregiver Portal, Explainable AI Recommendation Engine, Technical Documentation, Source Code Repository, Deployment Guide, Demo Video

18. [GGSIPU2618] AI-Powered Smart Classroom Attendance & Analytics System using Multi-Camera Computer Vision

Category: Software	**Theme:** Smart Education & Campus Automation
Organization: University School of Automation & Robotics (USAR), GGSIPU	**Domain:** Computer Vision, Edge AI, Artificial Intelligence (AI), Deep Learning, Face Recognition, Video Analytics, IoT, Edge Computing

Problem Statement:

University School of Automation & Robotics, Guru Gobind Singh Indraprastha University, East Delhi Campus runs four B.Tech programmes across large classrooms where attendance is still taken by roll-call or by circulating a sheet, consuming teaching time in every lecture, remaining susceptible to proxy attendance, and producing attendance data too late to act on the 75 percent detention rule before it is too late for the student. Classrooms already carry CCTV cameras that are used only for security. Develop an AI-powered Smart Classroom Attendance & Analytics System that automatically records attendance using multiple CCTV cameras installed inside classrooms while simultaneously providing occupancy monitoring, proxy detection, seat mapping, engagement analytics, classroom utilization statistics and real-time administrative dashboards. The system must identify students under varying lighting, occlusion, seating arrangement and camera viewpoint, integrate with the USAR timetable and enrolment records, support faculty verification and attendance dispute resolution, and enforce privacy, transparency and responsible AI — including student notification, consent, and a documented policy on retention of face data. (Example deployment: each USAR classroom is equipped with two CCTV cameras for complete coverage.)

Expected Solution:

Develop a multi-camera computer vision platform capable of processing synchronized video streams from classroom CCTV cameras using edge AI. The solution should perform face detection, face recognition, multi-camera identity association, multi-object tracking, attendance generation, proxy detection, classroom occupancy estimation, and seat mapping. The platform should integrate with institutional ERP/LMS systems to automatically map attendance with class schedules, faculty assignments, and enrolled students. It should provide administrator and faculty dashboards, attendance correction workflows, real-time alerts, absentee analytics, classroom utilization reports, engagement heatmaps, and historical attendance trends. The system should support privacy-preserving face embeddings, encrypted data storage, role-based access control, audit logs, and scalable deployment across an entire campus.

Technologies & Dataset:

Tech: Python, OpenCV, YOLOv11/YOLOv12, DeepFace, InsightFace, FaceNet, ByteTrack/DeepSORT, MediaPipe, TensorFlow, PyTorch, ONNX Runtime, FastAPI, Google Apps Script (Dashboard), Google Sheets/PostgreSQL, RTSP Streaming, Docker, GitHub | **Dataset:** CCTV Video Streams, Student Face Dataset (Institutional), Classroom Seating Plans, Timetable Data, ERP Student Records, Synthetic Classroom Video Dataset, Public Face Recognition Datasets (for research)

Deliverables:

Smart Attendance Engine, Multi-Camera Identity Tracking Module, Faculty Dashboard, Administrative Analytics Portal, Classroom Occupancy Dashboard, Attendance Reports, Proxy Detection Module, Classroom Utilization Analytics, Technical Documentation, Source Code Repository, Deployment Guide, Demo Video

19. [GGSIPU2619] AI-Based Automated Solar Filament Segmentation & Space Weather Intelligence System

Category: Software	**Theme:** Space Technology, Remote Sensing, Scientific AI
Organization: University School of Automation & Robotics (USAR), GGSIPU	**Domain:** Computer Vision, Deep Learning, Image Segmentation, Explainable AI, Scientific AI

Problem Statement:

Solar filaments are elongated magnetic structures suspended above the Sun's surface and act as important indicators of solar eruptions that may affect satellites, communication networks, navigation systems, and terrestrial power infrastructure. Manual identification and analysis of solar filaments from solar observation imagery is time-consuming and challenging due to their complex morphology, low contrast, fragmented appearance, overlapping structures, and varying observation conditions. Develop an AI-powered automated solar filament segmentation and intelligence system capable of detecting and segmenting individual solar filaments from multi-spectral solar imagery. The system should generate accurate pixel-level segmentation masks while preserving fine filament boundaries, reducing false detections, handling overlapping structures, and enabling large-scale solar monitoring for space weather prediction applications.

Expected Solution:

Develop an end-to-end AI pipeline for solar image preprocessing, noise reduction, contrast enhancement, automated filament segmentation, post-processing, and scientific analysis. The solution should utilize advanced semantic or instance segmentation models such as U-Net, DeepLabV3+, SegFormer, Mask2Former, Segment Anything Model (SAM), Vision Transformers, or hybrid Transformer-based architectures with attention mechanisms. The system should provide filament morphology analysis, confidence estimation, explainable AI visualizations, temporal evolution tracking, and efficient inference suitable for deployment in real-time space weather monitoring environments.

Technologies & Dataset:

Tech: Python, PyTorch, TensorFlow, OpenCV, CUDA, ONNX, Hugging Face Transformers, SAM, U-Net, DeepLabV3+, SegFormer, Mask2Former, Vision Transformer (ViT), MONAI, FastAPI, Docker | **Dataset:** Solar Dynamics Observatory (SDO) Data, H-Alpha Solar Images, GONG Solar Observatory Data, NSO Solar Data Archives, Solar Filament Benchmark Datasets, NASA Heliophysics Data Portal, ESA Solar Orbiter Data

Deliverables:

• Automated solar filament segmentation platform• Pixel-level filament segmentation masks• Solar image visualization dashboard with prediction overlays• Filament morphology and statistical analysis reports• Explainable AI module with attention maps/heatmaps• Trained deep learning models• Source code and documentation• Deployment guide

20. [GGSIPU2620] AI-Based 3D Microscopy Cell Detection, Tracking & Lineage Reconstruction System

Category: Software	**Theme:** Healthcare AI, Scientific Computing, Biomedical Image Analysis
Organization: University School of Automation & Robotics (USAR), GGSIPU	**Domain:** Computer Vision, Deep Learning, Medical Imaging, Object Tracking, Graph Neural Networks, Bioinformatics

Problem Statement:

Tracking and understanding cellular behavior over time using 3D microscopy is a critical challenge in modern biological research. Time-lapse volumetric microscopy generates large-scale 3D+time datasets where thousands of cells grow, migrate, interact, deform, and undergo division. Manual annotation and lineage reconstruction require significant expert effort and create a major bottleneck in biological discovery. Existing automated methods often fail due to dense cell populations, imaging noise, irregular cell morphology, occlusion, and complex division patterns. Develop an AI-powered system capable of detecting individual cells, tracking their movement across multiple time frames, identifying cell division events, and reconstructing complete cell lineage trees from 3D microscopy sequences. The solution should provide robust performance across diverse biological datasets and enable scalable analysis for developmental biology, immunology, disease modeling, and drug discovery applications.

Expected Solution:

Develop an end-to-end deep learning pipeline for 3D cell detection, temporal association, tracking, and lineage reconstruction. The system should perform volumetric image preprocessing, cell instance detection, feature extraction, multi-object tracking, and graph-based lineage generation. Advanced approaches may include 3D CNNs, Vision Transformers, Siamese networks, object tracking algorithms, graph neural networks, temporal transformers, and self-supervised learning methods. The solution should handle dense cell populations, noisy microscopy images, irregular cell shapes, and accurately identify mitosis/division events. The final output should generate structured cell graphs containing cell nodes, temporal positions, parent-child relationships, and lineage information.

Technologies & Dataset:

Tech: Python, PyTorch, TensorFlow, OpenCV, MONAI, CUDA, 3D CNNs, Vision Transformers, U-Net 3D, Mask R-CNN 3D, Segment Anything Model (SAM), Graph Neural Networks (GNN), Temporal Transformers, NumPy, SciPy, NetworkX, Docker, FastAPI | **Dataset:** 3D Time-Lapse Microscopy Datasets, Cell Tracking Challenge (CTC) Datasets, Synthetic Cell Lineage Datasets, Fluorescence Microscopy Data, Zarr-based Volumetric Imaging Datasets

Deliverables:

• Automated 3D cell detection system• Multi-frame cell tracking pipeline• Cell division detection module• Cell lineage reconstruction graph• Visualization dashboard for cell movement and genealogy• Prediction files containing cell nodes and temporal edges• Trained AI models and inference pipeline• Technical documentation and deployment guide

21. [GGSIPU2621] Exoplanet Atmospheric Spectrum Recovery from Noisy Telescope Observations

Category: Software	**Theme:** Space Technology, Astronomy AI, Scientific Machine Learning
Organization: University School of Automation & Robotics (USAR), GGSIPU	**Domain:** Deep Learning, Time-Series Analysis, Signal Processing, Astronomy Data Science, Uncertainty Modeling

Problem Statement:

Understanding the composition of exoplanet atmospheres is one of the major goals of modern astronomy. During a planetary transit, a small fraction of starlight passes through the planet's atmosphere, producing spectral signatures that reveal the presence of molecules such as water vapor, carbon dioxide, methane, and other chemical compounds. However, these signals are extremely weak and are often buried under complex time-dependent noise originating from stellar activity, telescope instruments, observation conditions, and detector imperfections. Develop an AI-powered system capable of recovering accurate exoplanet atmospheric spectra from noisy telescope observations. The solution should separate meaningful planetary signals from instrumental and stellar noise, generalize across different planetary systems, and estimate reliable uncertainty bounds for recovered spectra. The system should support future astronomical missions such as ESA's Ariel mission for large-scale exoplanet characterization.

Expected Solution:

Develop an end-to-end machine learning pipeline for astronomical signal reconstruction involving data preprocessing, noise characterization, spectral denoising, feature extraction, and uncertainty-aware prediction. The solution should leverage advanced deep learning architectures including Transformers, Neural Processes, Bayesian Neural Networks, Variational Autoencoders, Diffusion Models, CNN-LSTM architectures, and physics-informed machine learning approaches. Models should learn complex relationships between stellar observations, planetary atmospheric models, instrument characteristics, and observation cadence. The final system should generate predicted wavelength-wise spectra along with confidence intervals and uncertainty estimates suitable for scientific analysis.

Technologies & Dataset:

Tech: Python, PyTorch, TensorFlow, JAX, NumPy, SciPy, Pandas, Astropy, Transformers, Bayesian Neural Networks, Gaussian Processes, Diffusion Models, CNN-LSTM, Physics-Informed Neural Networks, CUDA, Docker | **Dataset:** ESA Ariel Mission Simulated Dataset, NeurIPS Ariel Data Challenge Dataset, Exoplanet Atmospheric Models, Synthetic Transit Spectroscopy Data, Stellar Noise Simulation Data, Telescope Instrument Calibration Data

Deliverables:

• AI-based exoplanet spectrum recovery system• Noise removal and signal reconstruction pipeline• Predicted atmospheric spectra with uncertainty estimates• Visualization dashboard for observed vs recovered spectra• Chemical composition analysis module• Trained deep learning models• Source code and documentation• Deployment and inference pipeline

22. [GGSIPU2622] Sleep State Detection and Sleep Pattern Analysis Using Wearable Sensor Data

Category: Software	**Theme:** Healthcare AI, Digital Health, Wearable Intelligence, Preventive Healthcare
Organization: University School of Automation & Robotics (USAR), GGSIPU	**Domain:** Time-Series Analysis, Machine Learning, Deep Learning, Signal Processing, Healthcare Analytics

Problem Statement:

Sleep plays a critical role in physical health, cognitive development, emotional regulation, and behavioral well-being. However, large-scale sleep research is limited due to the difficulty of collecting accurate sleep annotations in natural environments. Traditional methods such as sleep logs are subjective, time-consuming, and unable to accurately identify the transition between resting, falling asleep, and waking states. Wrist-worn accelerometers provide continuous physiological activity data but require advanced algorithms to accurately determine sleep and wake periods. Develop an AI-powered sleep monitoring system capable of detecting sleep onset, wake periods, and sleep patterns from wearable accelerometer data. The system should overcome challenges such as individual behavioral differences, noisy sensor measurements, irregular activity patterns, and large-scale population variability to support reliable sleep research and personalized health interventions.

Expected Solution:

Develop a machine learning pipeline that processes wrist-worn accelerometer time-series data to classify sleep and wake states. The solution should include signal preprocessing, activity feature extraction, temporal modeling, and sleep event detection. Advanced approaches may include deep learning models such as LSTM, Temporal CNNs, Transformers, Hidden Markov Models, Gradient Boosting, and hybrid sensor fusion techniques. The system should accurately detect sleep onset and wake transitions, estimate sleep duration, identify sleep disturbances, and provide personalized sleep analytics. The solution should be scalable for large population-level sleep studies and integration into digital health platforms.

Technologies & Dataset:

Tech: Python, PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, SciPy, Time-Series Models, LSTM, Temporal CNN, Transformers, XGBoost, Signal Processing Libraries, Wearable Sensor SDKs, FastAPI, Docker | **Dataset:** Healthy Brain Network (HBN) Dataset, Child Mind Institute Sleep Dataset, Wrist-Worn Accelerometer Data, Wearable Sensor Time-Series Data, Sleep Annotation Labels

Deliverables:

• AI-based sleep detection model• Sleep onset and wake classification system• Sleep pattern analytics dashboard• Individual sleep report generation• Visualization of activity and sleep cycles• Trained ML/DL models• Data preprocessing pipeline• Deployment documentation and API interface

23. [GGSIPU2623] Autonomous AI Agent for Strategic Farm Management and Dynamic Economic Optimization

Category: Software	**Theme:** Agentic AI, Autonomous Systems, Simulation Intelligence, Decision Automation
Organization: University School of Automation & Robotics (USAR), GGSIPU	**Domain:** Reinforcement Learning, Multi-Agent Systems, Large Language Models, Planning Algorithms, Optimization, Game AI

Problem Statement:

Real-world agricultural systems involve complex decision-making under uncertainty, including resource allocation, crop planning, livestock management, labor optimization, market fluctuations, and long-term investment strategies. Develop an autonomous AI farming agent capable of managing a virtual farm in a dynamic economic simulation environment. The agent should independently make strategic decisions including crop selection, planting schedules, irrigation, fertilization, harvesting, livestock management, land expansion, labor hiring, and market trading. The system should compete against other autonomous agents in a multi-agent environment while maximizing profitability over a simulated farming season. The challenge requires building intelligent systems capable of long-horizon planning, adaptation to changing market conditions, and efficient resource utilization similar to real-world agricultural supply chains.

Expected Solution:

Develop an autonomous AI agent capable of perceiving the current farm state, reasoning about future outcomes, and executing optimal actions through interaction with the simulation environment. The solution should combine reinforcement learning, planning algorithms, evolutionary optimization, Monte Carlo tree search, multi-agent learning, and LLM-based decision-making approaches. The agent should learn optimal strategies for crop production, animal management, labor allocation, market prediction, and investment decisions. Advanced solutions may include hierarchical agents with specialized modules for finance, agriculture planning, resource management, and competitive strategy. The final system should continuously adapt to economic changes and optimize long-term farm profitability.

Technologies & Dataset:

Tech: Python, Reinforcement Learning Frameworks, PyTorch, TensorFlow, OpenAI Gym/Gymnasium, Stable-Baselines3, Deep Q-Networks (DQN), PPO, Actor-Critic Models, Monte Carlo Tree Search, LLM Agents, LangChain/LangGraph, Optimization Algorithms, Simulation APIs | **Dataset:** Virtual Farm Simulation Environment, Dynamic Market Data, Crop Growth Simulation Data, Resource Management Logs, Agent Interaction History, Economic State Transitions

Deliverables:

• Autonomous farming AI agent• Decision-making and planning engine• Market prediction module• Resource optimization system• Multi-agent competition framework• Farm performance analytics dashboard• Trained agent models• Simulation integration code• Technical documentation

24. [GGSIPU2624] Human Preference Prediction Model for Large Language Model Responses

Category: Software	**Theme:** Generative AI, Responsible AI, Human-AI Alignment, RLHF
Organization: University School of Automation & Robotics (USAR), GGSIPU	**Domain:** Natural Language Processing, Large Language Models, Preference Modeling, Reinforcement Learning from Human Feedback

Problem Statement:

Large Language Models (LLMs) are increasingly becoming part of everyday applications, but generating technically correct responses alone is insufficient for effective human interaction. AI systems must produce responses that align with human expectations, preferences, and communication styles. Existing evaluation approaches often rely on automated metrics or direct model-based judging, which may introduce biases such as response order preference, verbosity bias, or model self-preference. Develop an AI-powered preference prediction system capable of predicting which response a human user would prefer when comparing two competing LLM-generated answers. The system should learn from real-world human feedback data and act as a reward model for improving conversational AI alignment. The solution should capture factors such as helpfulness, relevance, correctness, clarity, safety, and user satisfaction.

Expected Solution:

Develop a machine learning model that predicts human preference between two LLM responses given the same user prompt. The system should utilize advanced NLP architectures including transformer-based encoders, instruction-tuned language models, pairwise ranking models, reward models, and preference optimization techniques. The solution may incorporate methods such as Reinforcement Learning from Human Feedback (RLHF), Direct Preference Optimization (DPO), Contrastive Learning, Siamese Networks, and LLM-based feature extraction. The model should identify subtle response quality differences while minimizing biases such as position bias, verbosity bias, and stylistic preference bias. The final system should provide preference scores or rankings for competing AI-generated responses.

Technologies & Dataset:

Tech: Python, PyTorch, TensorFlow, Hugging Face Transformers, LLaMA, BERT, DeBERTa, RoBERTa, Reward Modeling Frameworks, RLHF, DPO, Contrastive Learning, Sentence Transformers, NLP Libraries, CUDA, MLflow, FastAPI | **Dataset:** Chatbot Arena Human Preference Dataset, LLM Battle Logs, Pairwise Response Comparison Dataset, User Feedback Data, Human Ranking Annotations

Deliverables:

• Human preference prediction model• LLM reward model• Response ranking API• Preference analysis dashboard• Bias detection and mitigation module• Model evaluation framework• Trained NLP models• Source code and documentation

25. [GGSIPU2625] Kolam Pattern Understanding and Generative Reconstruction System

Category: Software	**Theme:** Cultural Heritage, Generative AI, Pattern Recognition, Digital Preservation
Organization: University School of Automation & Robotics (USAR), GGSIPU	**Domain:** Computer Vision, Image Processing, Generative Algorithms, Computational Geometry, Artificial Intelligence

Problem Statement:

Kolam is a traditional Indian geometric art form created using mathematical patterns, symmetry, and repetitive structures. These designs contain complex rules involving dots, curves, loops, reflections, rotations, and spatial relationships. Develop computer programs capable of identifying the underlying design principles of Kolam patterns and automatically recreating these traditional artworks. The solution should analyze existing Kolam images, understand their geometric structure, symmetry, and construction rules, and generate accurate digital reproductions while preserving traditional artistic characteristics. The system should contribute to computational understanding, preservation, and digital generation of cultural designs.

Expected Solution:

Develop an AI-powered Kolam analysis and generation framework that detects structural patterns, identifies geometric rules, and reconstructs Kolam designs algorithmically. The solution may use computer vision techniques for pattern extraction, image segmentation, symmetry detection, graph-based representations, procedural generation algorithms, and generative AI models. The system should identify elements such as dot grids, line connectivity, rotational/reflection symmetry, and curve formations before generating new or reconstructed Kolam designs. Advanced solutions may include interactive Kolam generation tools where users can modify parameters and generate culturally consistent patterns.

Technologies & Dataset:

Tech: Python, OpenCV, NumPy, Matplotlib, Scikit-image, Computer Vision Algorithms, Generative Algorithms, Computational Geometry, Graph Theory, GANs, Diffusion Models, Neural Networks, Image Processing Libraries | **Dataset:** Traditional Kolam Image Dataset, Digitized Cultural Art Archives, Hand-Drawn Kolam Images, Synthetic Geometric Pattern Dataset

Deliverables:

• Kolam image analysis system• Pattern recognition module• Automated Kolam reconstruction engine• Procedural Kolam generator• Symmetry and geometry analysis module• Visualization interface• Source code and documentation• Digital archive of generated designs

26. [GGSIPU2626] Blockchain-Based Traceability System for Ayurvedic Herb Supply Chain and Authentication

Category: Software	**Theme:** Blockchain, Healthcare Technology, Traditional Medicine, Supply Chain Transparency
Organization: University School of Automation & Robotics (USAR), GGSIPU	**Domain:** Blockchain, IoT, Geospatial Technology, Data Integrity, Supply Chain Management, Digital Authentication

Problem Statement:

Ayurvedic medicines rely on medicinal plants collected from farms, forests and traditional sources, yet lack of supply chain transparency, adulteration, incorrect botanical identification, uncertain geographical origin and limited traceability from collection to formulation undermine authenticity and consumer trust. GGSIPU is well placed to address this as a university: its affiliated Ayurveda and health-sciences institutions supply the domain expertise in botanical identification, pharmacognosy and formulation, while University School of Automation & Robotics, East Delhi Campus supplies the blockchain, IoT and geospatial engineering capability making this a genuine cross-disciplinary university problem rather than a purely technical one. Develop a blockchain-based botanical traceability system tracking Ayurvedic herbs across the complete lifecycle from collection by farmers or wild collectors through processing, formulation, packaging and final labelling, maintaining tamper-proof records, capturing geo-location at source, verifying authenticity and providing transparent access to collectors, manufacturers, regulators and consumers.

Expected Solution:

Develop a decentralized traceability platform using blockchain technology to record every stage of the Ayurvedic herb supply chain. The solution should include geo-tagged collection records, digital identity management for farmers and collectors, batch-level tracking, smart contracts for supply chain verification, and QR-based consumer authentication. IoT-enabled sensors may be integrated for environmental monitoring, storage condition tracking, and quality assurance. The system should ensure immutability of records, prevent counterfeit products, and enable end-to-end visibility from botanical source to finished Ayurvedic formulation.

Technologies & Dataset:

Tech: Blockchain (Ethereum/Hyperledger Fabric), Smart Contracts, Solidity, Web3 APIs, IPFS, IoT Sensors, GPS/GIS Technologies, Cloud Platforms, Mobile Applications, QR Code Systems, Databases (PostgreSQL/MongoDB), REST APIs, Cryptography | **Dataset:** Ayurvedic Herb Databases, Botanical Species Records, Geographic Collection Data, Farmer/Collector Supply Chain Records, Herbal Product Batch Data, Quality Certification Records

Deliverables:

• Blockchain-based herb traceability platform• Farmer/collector registration module• Geo-tagged botanical collection system• Smart contract-based supply chain tracking• QR-code product authentication system• Consumer verification portal• Supply chain analytics dashboard• APIs and deployment documentation

27. [GGSIPU2627] AI-Based Timetable Generation System Aligned with NEP 2020 for Multidisciplinary Education Structures

Category: Software	**Theme:** Education Technology, AI Automation, NEP 2020 Implementation, Smart Campus
Organization: TimeTable Committee, University School of Automation & Robotics (USAR), GGSIPU	**Domain:** Artificial Intelligence, Constraint Optimization, Scheduling Algorithms, Machine Learning, Decision Support Systems

Problem Statement:

NEP 2020 introduces multidisciplinary structures, flexible course choices, multiple entry-exit options, interdisciplinary electives, skill-based courses and increased academic flexibility, all of which various University Schools located at Dwarka Campus as well as East Delhi Campus are now implementing. At USAR, this is concrete and difficult: four B.Tech programmes share a limited pool of classrooms, specialised laboratories, workshops and faculty, students select Program Core Electives (PCE) and Open Area Electives (OAE) across programmes, laboratory batches must be split and rotated, guest and visiting faculty are available only on fixed days, and every change to one section cascades through the rest. Manual timetable preparation each semester consumes weeks of senior faculty time and still yields clashes discovered only after classes begin. Develop an AI-based automated timetable generation system capable of producing optimized, conflict-free academic schedules aligned with NEP 2020 principles, supporting diverse programmes, flexible curricula, interdisciplinary course selection and institutional constraints while minimising conflicts and maximising resource utilization.

Expected Solution:

Develop an intelligent timetable generation framework using AI and optimization techniques to automatically generate conflict-free academic schedules. The solution should model timetable generation as a constraint satisfaction and optimization problem using approaches such as Genetic Algorithms, Reinforcement Learning, Constraint Programming, Integer Linear Programming, and Hybrid AI optimization. The system should consider faculty workload, classroom and laboratory availability, student course combinations, elective structures, interdisciplinary courses, academic regulations, and NEP 2020 curriculum flexibility. The platform should provide automated generation, conflict detection, modification suggestions, and dynamic timetable updates.

Technologies & Dataset:

Tech: Python, Machine Learning, Optimization Algorithms, Genetic Algorithms, Reinforcement Learning, Constraint Programming, OR-Tools, Integer Linear Programming, Django/FastAPI, PostgreSQL/MySQL, React/Angular, Cloud Deployment, AI Planning Algorithms | **Dataset:** University Academic Data, Course Catalog, Faculty Information, Classroom/Lab Resources, Student Enrollment Data, NEP 2020 Curriculum Structures, Academic Regulations

Deliverables:

• AI-powered timetable generation platform• Constraint optimization engine• Faculty workload management module• Classroom and resource allocation module• Conflict detection and resolution system• Admin dashboard• Student/faculty timetable portal• Automated report generation• Deployment documentation

28. [GGSIPU2628] Digital Farm Management Portal for Monitoring Maximum Residue Limits (MRL) and Antimicrobial Usage (AMU) in Livestock

Category: Software	**Theme:** Smart Farming, Food Safety, Digital Agriculture, Animal Health Monitoring
Organization: University School of Automation & Robotics (USAR), GGSIPU	**Domain:** IoT, Data Analytics, Healthcare Informatics, Farm Management Systems, Regulatory Compliance

Problem Statement:

Excessive and improper use of antimicrobials in livestock can lead to antimicrobial resistance and unsafe levels of drug residues in animal-derived food products. Monitoring Antimicrobial Usage (AMU) and ensuring compliance with Maximum Residue Limits (MRL) are critical for food safety, public health, and sustainable livestock production. Develop a digital farm management portal capable of recording, monitoring, and analyzing livestock health data, antimicrobial administration, withdrawal periods, and residue compliance. The system should enable farmers, veterinarians, regulators, and food supply chain stakeholders to track medicine usage, detect potential MRL violations, and improve responsible antimicrobial practices in livestock farming.

Expected Solution:

Develop an integrated digital livestock management platform that captures animal records, medication history, veterinary prescriptions, antimicrobial usage patterns, and laboratory residue testing data. The solution should provide automated compliance monitoring against MRL standards, alerts for unsafe antimicrobial usage, analytics dashboards, and predictive insights for disease management. Advanced solutions may incorporate IoT-based animal monitoring, AI-driven disease prediction, blockchain-based medicine traceability, and mobile applications for farmers. The platform should support real-time monitoring and evidence-based decision-making for sustainable livestock management.

Technologies & Dataset:

Tech: Python, Machine Learning, IoT Sensors, Cloud Computing, Blockchain, Mobile Application Development, FastAPI/Django, PostgreSQL/MongoDB, Data Analytics, AI Prediction Models, GIS Technologies, QR Code Systems | **Dataset:** Livestock Health Records, Veterinary Prescription Data, Antimicrobial Usage Logs, Residue Testing Data, Farm Management Records, Food Safety Standards and MRL Guidelines

Deliverables:

• Digital livestock farm management portal• Animal health record management system• AMU tracking module• MRL compliance monitoring engine• Automated alert and notification system• Veterinary dashboard• Farmer mobile interface• Analytics and reporting module• API documentation and deployment guide

29. [GGSIPU2629] AI-Assisted Building Inspection & Structural Health Monitoring with Defect Detection and Compliance Reporting

Category: Software	**Theme:** Smart Infrastructure & Inspection
Organization: University School of Automation & Robotics (USAR), GGSIPU	**Domain:** Computer Vision, Structural Health Monitoring, Deep Learning, NLP, Construction Compliance, Drone Image Analysis

Problem Statement:

University School of Autonation & Robotics has the technical expertise and campus infrastructure to pioneer AI-driven building inspection and structural health monitoring. To advance this innovative domain, develop a computer vision platform that processes ground-level and drone-captured imagery to detect structural defects and ensure regulatory compliance. Rather than relying on manual visual examination, which can be inconsistent and delayed in identifying defects, an AI-assisted platform will generate quantified defect maps, structured reports, and audit trails for inspectors and authorities. This solution will enhance campus safety, support statutory compliance workflows, and demonstrate institutional innovation in smart infrastructure. Develop an AI-assisted building inspection platform that processes ground-level and drone-captured images and video to detect and localize defects such as cracks, corrosion, water seepage, spalling, unauthorized structural additions, and regulatory violations, and generates structured inspection reports through NLP. The platform should schedule statutory inspections, track compliance with building codes, and support campus building inspection workflows, offering quantitative defect maps and confidence-scored findings for auditors.

Expected Solution:

Develop a computer-vision pipeline (instance segmentation and anomaly detection) over drone and mobile imagery with defect localization, severity scoring, and confidence estimates; an NLP report generator producing structured, code-referenced inspection reports; an inspection scheduling and compliance tracker; and role-based dashboards for inspectors and authorities. The system should provide APIs for importing drone footage and exporting inspection reports.

Technologies & Dataset:

Tech: PyTorch, YOLO/SegFormer/Detectron2, OpenCV, FastAPI, React, PostgreSQL/PostGIS, GeoTIFF/GeoJSON, Transformers (NLP), Docker, GitHub | **Dataset:** Public Defect Detection Datasets (e.g., SDNET Concrete Crack Data), Sample Drone Imagery, Building Floor Plans, Compliance Checklist Templates, User-Collected Campus Images

Deliverables:

Inspection Image Ingestion Module, Defect Detection & Localization Engine, Severity Scoring Module, NLP Report Generator, Inspection Scheduling & Compliance Tracker, Inspector & Authority Dashboards, API Documentation, Technical Documentation, Source Code Repository, Deployment Guide, Demo Video

30. [GGSIPU2630] AI-Powered Smart Campus Parking Management with ANPR, Permit Tiers, and Dynamic Slot Allocation

Category: Software	**Theme:** Smart Campus Infrastructure & IoT
Organization: University School of Automation & Robotics (USAR), GGSIPU	**Domain:** Computer Vision, ANPR, License Plate Recognition, Parking Optimization, IoT, Mobile Application Development, Data Analytics

Problem Statement:

As GGSIPU East Delhi Campuscontinues to grow as a smart campus, implementing an intelligent parking management system will enhance operational efficiency, enforce equitable permit policies, and improve the student and faculty experience. By deploying ANPR-based gate barriers and real-time slot detection, the institution can transition from unmanaged parking access to data-driven allocation and violation enforcement. This platform will serve as a showcase of IoT and AI integration for a campus environment while addressing practical parking challenges specific to Indian traffic conditions. Develop a smart parking system for campus environments featuring: (1) ANPR-based gate barriers that recognize number plates for automatic entry and exit and enforce permit tiers (faculty, student, staff, visitor, EV); (2) real-time slot availability detection using cameras and/or sensors with dynamic guidance; (3) a booking and reservation system including EV-charging bay reservation; (4) violation and overstay enforcement with automated notifications; (5) occupancy analytics and reports for authorities; and (6) a mobile app for permits, bookings, payments, and live availability. The solution must work reliably under Indian conditions, including diverse plate fonts, monsoon lighting, and mixed two-wheeler and four-wheeler traffic.

Expected Solution:

Develop an edge-deployed ANPR pipeline (plate detection and multi-font OCR) integrated with gate barrier controllers via APIs, slot-occupancy detection through CCTV cameras, a reservation and booking engine with permit tiers and EV-bay priority, fine and violation management, and analytics dashboards. The system should provide a mobile-first interface for users and a web dashboard for parking authorities, with MQTT-based IoT integration where sensors are deployed.

Technologies & Dataset:

Tech: PyTorch/TensorFlow, PaddleOCR/Tesseract, OpenCV, FastAPI, React Native/Flutter, PostgreSQL/Redis, MQTT (IoT), JWT Authentication, Docker, GitHub | **Dataset:** Public Vehicle Number-Plate Datasets (e.g., CCPD, Indian Plate Collections), Synthetic Campus Traffic Logs, Parking Layout Data, Permit Records, EV Charging Bay Data

Deliverables:

ANPR & Plate Recognition Module, Gate Barrier Integration, Slot Occupancy Detection Engine, Booking & Reservation Engine with EV Bay Priority, Permit & Violation Management Module, Mobile App, Authority Analytics Dashboard, Technical Documentation, Source Code Repository, Deployment Guide, Demo Video

31. [GGSIPU2631] Training & Placement Cell (TPC) Digital Workflow Platform: Drive Management, Eligibility Gating, Offer Policy Enforcement & NIRF Reporting

Category: Software	**Theme:** Placement & Career Services Automation
Organization: University School of Automation & Robotics (USAR), GGSIPU	**Domain:** Placement Management, Eligibility & Constraint Engines, Workflow Automation, Analytics, Resume & Document Management, Report Generation

Problem Statement:

Training and Placement Cell plays a vital role in student career outcomes and institutional rankings. To enhance placement operations and streamline the TPO workflow, an integrated digital platform can centralize drive management, ensure transparent eligibility verification, and automate compliance with placement policies. While placement cells currently coordinate through spreadsheets and manual verification, a structured workflow platform will improve data accuracy, reduce administrative burden on TPO staff, and enable precise NIRF placement reporting. This platform will strengthen GGSIPU's competitive advantage in placement metrics and student satisfaction. Develop a TPC platform for the placement office that manages: (1) a drive calendar with company onboarding and drive configuration; (2) automated eligibility gating against CGPA, backlog, branch, and year rules; (3) student application management with a resume bank; (4) offer and rejection lifecycle tracking including one-offer-one-student and dream-offer policies with configurable constraints; (5) training program scheduling (aptitude, soft skills, certifications) with participation tracking; (6) analytics and NIRF placement reports (offers, packages, salary bands, branch-wise counts) with one-click export; and (7) role-based portals for TPO, companies, faculty coordinators, and students.

Expected Solution:

Develop a workflow-driven placement platform with configurable drive templates, an eligibility engine, an offer-policy constraint engine, resume and application management, a training module with participation tracking, and a report generator producing placement statistics and NIRF-format exports. The system should provide automated notifications, role-based dashboards for TPO, companies, faculty coordinators, and students, and integrate with existing student and attendance records.

Technologies & Dataset:

Tech: React/Next.js, Django/FastAPI, PostgreSQL, Celery/Redis, Excel/PDF Export Libraries, Email Gateway, JWT Authentication, Bootstrap/Tailwind CSS, Docker, GitHub | **Dataset:** Sample Drive Records, Academic Records (CGPA, Backlogs), Student Resume Bank, Offer & Rejection Records, Training Program Data, NIRF Reporting Templates, Synthetic Test Data

Deliverables:

Drive Calendar & Configuration Module, Eligibility Gating Engine, Application & Resume Management Module, Offer Lifecycle & Policy Enforcement Engine, Training Program Module, NIRF Report Generator, TPO & Company & Student Portals, Technical Documentation, Source Code Repository, Deployment Guide, Demo Video

32. [GGSIPU2632] University Research & Funding Lifecycle Management: Grants, Utilization Certificates & GFR-Compliant Expenditure Tracking

Category: Software	**Theme:** Research Administration & Financial Compliance
Organization: University School of Automation & Robotics (USAR), GGSIPU	**Domain:** Grants Management, Financial Compliance (GFR), Utilization Certificates, Fund Monitoring, Workflow Automation, Analytics

Problem Statement:

GGSIPU's growing research portfolio spanning research grants, CSR initiatives, endowment funds, and alumni donations requires robust financial governance to ensure GFR compliance and timely utilization certificate generation. To modernize research administration and reduce the audit preparation burden, a dedicated funding lifecycle platform can centralize grant registration, expenditure tracking, and compliance documentation. Currently, fund management relies on spreadsheets and manual registers, which can delay utilization certificates and complicate audits; a unified system will strengthen institutional credibility with sponsors and funding agencies. Develop a university funding lifecycle platform covering: (1) grant and project registration with sanctioned amounts, budget heads, and timelines; (2) expenditure entry and tracking against heads with GFR-based rules and deviation alerts; (3) automatic generation of utilization certificates and auditable statements; (4) milestone and progress reporting; (5) donor, sponsor, and funding-agency portals; (6) alumni donation and endowment or corpus tracking; and (7) consolidated fund-utilization dashboards for finance, HoDs, and administration.

Expected Solution:

Develop a web-based platform with a budget and expenditure ledger engine supporting head-wise balances, GFR rule checks, and committed-versus-spent tracking; automatic generation of utilization certificates and financial statements; milestone tracking; role-based portals for principal investigators, finance, sponsors, and alumni; and consolidated analytics dashboards. The system should provide exportable audit statements and integrate with the existing notesheet generation system for sanctions and approvals.

Technologies & Dataset:

Tech: React/Next.js, Django, PostgreSQL, Celery/Redis, PDF Generation, Excel Export, JWT Authentication, Bootstrap/Tailwind CSS, Docker, GitHub | **Dataset:** Synthetic Grant Sanctions & Budget Heads, Expenditure Records, GFR Templates, Sample Utilization Certificate Formats, Donor & Endowment Data, Milestone Reports

Deliverables:

Grant & Project Registration Module, Budget & Expenditure Ledger Engine, GFR Rule Check Module, Utilization Certificate Generator, Milestone Tracking Module, Sponsor & Donor & Alumni Portals, Fund-Utilization Dashboards, Technical Documentation, Source Code Repository, Deployment Guide, Demo Video

33. [GGSIPU2633] Disaster Relief Resource-Demand Matching & Logistics Coordination Platform

Category: Software	**Theme:** Disaster Management & Response
Organization: University School of Automation & Robotics (USAR), GGSIPU	**Domain:** Disaster Response, Resource Optimization, Matching & Routing Algorithms, Crowdsourcing, Geospatial Mapping, Mobile Application Development

Problem Statement:

University School of Automation & Robotics expertise in technology and optimization can contribute meaningfully to disaster resilience and humanitarian response. During natural disasters such as floods, earthquakes, and cyclones, relief agencies, volunteers, and citizens lack a coordinated view of relief needs, available resources, and optimal distribution routes. Relief requests and donations are currently coordinated through fragmented channels phone calls, social media, and informal networks which can cause duplication, delays, and inefficient distribution. A technology-enabled coordination platform will strengthen relief ecosystem efficiency and demonstrate institutional commitment to societal impact. Develop a disaster relief coordination platform that: (1) registers relief resources (shelters, food, water, medicines, rescue teams, vehicles) with location, capacity, and status; (2) ingests demand requests from agencies and crowdsourced channels; (3) runs a resource-demand matching and allocation engine with priority scoring and dispatch recommendations; (4) provides routing guidance for delivery vehicles; (5) offers a public view for affected persons showing nearby shelters and relief points; and (6) aggregates situational information for responders with post-disaster reporting.

Expected Solution:

Develop a map-based coordination platform with a relief resource registry, demand intake through forms and social/API ingestion, a matching and optimization engine with dispatch recommendations, vehicle routing support, a public shelter locator, and reporting dashboards. The system should include an offline-tolerant mobile application for ground teams and integrate spatial data for mapping, with role-based access for relief agencies, volunteers, and administrators.

Technologies & Dataset:

Tech: React, Django/FastAPI, PostgreSQL/PostGIS, Google OR-Tools (Routing), WebSocket, Flutter (Mobile), MQTT, OpenStreetMap Integration, Docker, GitHub | **Dataset:** Synthetic Disaster Demand & Resource Datasets, OpenStreetMap Spatial Data, Historical NDMA/NDRF Datasets, Crowdsourced Reports (Sample), Relief Agency Records

Deliverables:

Relief Resource Registry, Demand Intake Module (Forms & API), Resource-Demand Matching Engine, Vehicle Routing Module, Public Shelter Locator, Situational Awareness Dashboard, Offline-Tolerant Mobile App, Technical Documentation, Source Code Repository, Deployment Guide, Demo Video

34. [GGSIPU2634] Institution-Facing Industry Demand & Curriculum Gap Analysis Platform (Skills to Courses)

Category: Software	**Theme:** Industry-Academia Alignment
Organization: University School of Automation & Robotics (USAR), GGSIPU	**Domain:** NLP, Labour Market Analytics, Skill Taxonomy Mapping, Curriculum Analytics, Data Extraction & Processing, Dashboards & Reporting

Problem Statement:

University School of Automation & Robotics is committed to ensuring that its curricula remain aligned with industry demand and equip students with market-relevant skills. To bridge the information gap between emerging industry needs and academic program design, an institution-facing analytics platform can process real-time job market signals and compare them against course offerings. While individual career platforms match students to jobs, this solution focuses on enabling academic leadership deans and HoDs to make data-informed curriculum updates. Develop a platform that: (1) ingests live job postings from the National Career Service (NCS), job portals, and company career pages and extracts in-demand skills, roles, and salary bands through NLP; (2) maps program outcomes and course syllabi to standard skill taxonomies; (3) computes curriculum-industry gap scores for each program and branch; (4) generates reports and dashboards for deans and HoDs showing opportunities to modernize courses and add emerging skill electives; (5) tracks graduate skill attainment against market demand; and (6) surfaces actionable recommendations to update curricula and training programs.

Expected Solution:

Develop an NLP pipeline for job-posting ingestion, skill extraction, and taxonomy mapping, integrated with a curriculum repository and a gap-scoring engine. The system should provide dashboards and periodic reports for academic leadership, exportable recommendation briefs for curriculum revision, and APIs for plugging in additional job sources and university ERP data. Skill attainment tracking should leverage existing attendance, competency, and academic records where available.

Technologies & Dataset:

Tech: Python, FastAPI, Transformers/NER Models, Elasticsearch, React, PostgreSQL, Celery/Redis, Docker, GitHub | **Dataset:** Public NCS/Job-Portal Sample Postings, O*NET/Skill Taxonomies, Course Syllabi (Institution-Provided), Synthetic Graduate Outcome Data, Labour Market Reports

Deliverables:

Job Posting Ingestion Module, NLP Skill Extraction Engine, Skill Taxonomy Mapping Module, Curriculum Repository, Gap-Scoring Engine, Dean/HoD Dashboards, Recommendation Brief Exporter, API Documentation, Technical Documentation, Source Code Repository, Deployment Guide, Demo Video

35. [GGSIPU2635] Faculty Appraisal, API Score Automation & NAAC-NIRF Data Pipeline

Category: Software	**Theme:** Academic Governance & Accreditation
Organization: University School of Automation & Robotics (USAR), GGSIPU	**Domain:** Faculty Evaluation, Academic Performance Indicators (API), Publications Analytics, NAAC/NIRF Reporting, Workflow Automation, Data Integration

Problem Statement:

Guru Gobind Singh Indraprastha University pursuit of academic excellence and institutional rankings requires efficient, accurate faculty appraisal and accreditation reporting. To streamline the annual API scoring cycle and NAAC/NIRF data assembly, a dedicated platform can automate publication tracking, compute performance metrics consistently, and reduce the manual data compilation that currently spans several months. While a monolithic university ERP would disrupt existing systems, this focused solution addresses the critical governance gap integrating scattered records of faculty achievements into auditable appraisal and ranking data. Develop a platform that: (1) provides faculty self-appraisal with evidence upload; (2) auto-computes API scores and category-wise performance metrics from publications (Scopus, Web of Science, Crossref), patents, funded projects, and teaching load; (3) tracks approvals through role-based workflows (HoD, Dean, Administration); (4) assembles NAAC SSR and NIRF ranking data with versioned data collection; (5) generates dashboards and exportable reports following the UGC API scoring matrix and NAAC/NIRF formats; and (6) integrates with existing certificate, notesheet, and attendance modules where relevant.

Expected Solution:

Develop a web-based platform with a publication and project ingestion engine using DOI and Crossref APIs, an API-score calculator implementing UGC career-advancement matrix rules, a faculty appraisal workflow with role-based approvals, NAAC SSR and NIRF data assembly with audit trail, and a report generator. The system should provide dashboards for faculty and administration, automated publication imports, and exportable reports in NAAC/NIRF formats.

Technologies & Dataset:

Tech: React/Next.js, Django/FastAPI, PostgreSQL, Crossref/Scopus APIs, Celery/Redis, Excel/PDF Export, JWT Authentication, Bootstrap/Tailwind CSS, Docker, GitHub | **Dataset:** Sample Faculty Publication Records (via Crossref), Synthetic Appraisal & Project Data, UGC API Matrix Templates, NAAC SSR Formats, NIRF Data Formats, Committee Service Records

Deliverables:

Faculty Self-Appraisal Module, Publication/Project Ingestion Engine, API Score Calculator, Role-Based Approval Workflow, NAAC SSR Data Assembler, NIRF Report Generator, Faculty & Administration Dashboards, Technical Documentation, Source Code Repository, Deployment Guide, Demo Video

36. [GGSIPU2636] Low-Cost Smart Transportation Solution for Agri Produce from Remote Farms to Nearest Motorable Road in NER Region

Category: Hardware	**Theme:** Smart Transportation
Organization: University School of Automation & Robotics (USAR), GGSIPU	**Domain:** Transportation & Logistics

Problem Statement:

Develop prefabricated modular road panels using bamboo reinforcement and plastic waste-infused composites for durable, erosion-resistant roads in remote and hilly regions. The solution should address poor connectivity, difficult terrain, heavy rainfall, and limited conventional construction resources in the North Eastern Region.

Expected Solution:

Prototype modular road panel system using bamboo mesh/strips and recycled LDPE/HDPE plastic composites. Include drainage, slope adaptation, anti-slip surfaces, deployment plan for 100–200 m rural stretches, cost-benefit analysis, environmental impact assessment, and optional IoT monitoring.

Technologies & Dataset:

Tech: Embedded Systems, IoT Sensors, Composite Materials, Bamboo Engineering, Plastic Recycling Technology, Structural Design, GIS | **Dataset:** Terrain data, rainfall data, soil characteristics, road condition datasets (to be collected/generated)

Deliverables:

Functional prototype, deployment guide, cost analysis report, environmental assessment, scalability roadmap, optional IoT monitoring system

37. [GGSIPU2637] Waste Segregation Monitoring System for Urban Local Bodies

Category: Hardware	**Theme:** Clean & Green Technology
Organization: University School of Automation & Robotics (USAR), GGSIPU	**Domain:** Waste Management

Problem Statement:

Urban waste segregation at household level remains inefficient due to poor compliance and lack of monitoring mechanisms, increasing landfill dependency and recycling costs.

Expected Solution:

IoT-based waste segregation monitoring system with sensors and mobile application to track segregation behaviour, provide feedback to households, and generate compliance reports for authorities.

Technologies & Dataset:

Tech: IoT, Computer Vision, Machine Learning, Sensors, Mobile Application, Cloud Analytics | **Dataset:** Waste classification datasets, household waste image datasets, municipal waste records

Deliverables:

Prototype device, mobile application, monitoring dashboard, compliance reports

38. [GGSIPU2638] Intelligent Pesticide Sprinkling System Determined by Infection Level of a Plant

Category: Hardware	**Theme:** Agriculture, FoodTech & Rural Development
Organization: University School of Automation & Robotics (USAR), GGSIPU	**Domain:** Precision Agriculture

Problem Statement:

Excessive pesticide usage causes soil degradation, water contamination, and ecological damage. Traditional spraying applies chemicals uniformly without considering plant health status.

Expected Solution:

AI-enabled smart spraying system using cameras and sensors to detect plant diseases and regulate pesticide quantity based on infection severity.

Technologies & Dataset:

Tech: Computer Vision, Deep Learning, IoT, Edge AI, Embedded Systems, Smart Actuators | **Dataset:** Plant disease image datasets, crop health datasets

Deliverables:

Automated pesticide spraying prototype, farmer monitoring dashboard, AI disease detection model

39. [GGSIPU2639] Development of Indigenous Contactless Integrated Track Monitoring System (ITMS) for Track Recording on Indian Railways

Category: Hardware	**Theme:** Smart Automation
Organization: University School of Automation & Robotics (USAR), GGSIPU	**Domain:** Railway Infrastructure Monitoring

Problem Statement:

Develop indigenous contactless track monitoring technology replacing expensive imported laser-based systems. The system must monitor track geometry, rail profile, wear, acceleration, infringement, and track components.

Expected Solution:

Complete ITMS prototype complying with RDSO TM/IM/448 Rev.1:2023 and EN 13848 standards with real-time acquisition, processing, chainage mapping, and reporting.

Technologies & Dataset:

Tech: LiDAR/Laser Sensors, Computer Vision, IoT, Edge Computing, AI Signal Processing, Embedded Systems | **Dataset:** Railway track geometry datasets, sensor recordings, vibration datasets

Deliverables:

Hardware prototype, software platform, validation report, field demonstration video, technical documentation

40. [GGSIPU2640] AI-Based Development of Laser-Based QR Code Marking on Track Fittings on Indian Railways

Category: Hardware	**Theme:** Transportation & Logistics
Organization: University School of Automation & Robotics (USAR), GGSIPU	**Domain:** Railway Asset Management

Problem Statement:

Lack of identification and lifecycle tracking mechanism for rail fittings such as clips, liners, pads, and sleepers causes challenges in quality monitoring and inventory management.

Expected Solution:

Laser-marked QR identification system integrated with UDM and TMS portals. AI-based extraction of vendor details, warranty, inspection history, and performance reports through mobile scanning.

Technologies & Dataset:

Tech: Laser Marking, QR Technology, AI OCR, Mobile Computing, Blockchain, Cloud Database | **Dataset:** Railway asset records, fitting lifecycle data

Deliverables:

QR marking prototype, mobile scanning application, AI reporting dashboard, integration framework

41. [GGSIPU2641] E-Tongue for Dravya Identification

Category: Hardware	**Theme:** MedTech / BioTech / HealthTech
Organization: University School of Automation & Robotics (USAR), GGSIPU	**Domain:** Herbal Authentication

Problem Statement:

Traditional Ayurvedic herb identification relies on subjective taste assessment. Variations in human perception create inconsistency in quality assessment and authentication.

Expected Solution:

Sensor-integrated AI electronic tongue system capable of identifying herbs based on taste signatures and phytochemical profiles.

Technologies & Dataset:

Tech: Electronic Sensors, NIR/Raman Spectroscopy, AI/ML, Chemometrics, Neural Networks, Cloud Database | **Dataset:** Authenticated herb samples, phytochemical datasets, taste fingerprint database

Deliverables:

Portable e-tongue prototype, ML classification model, herb database, adulteration detection system

42. [GGSIPU2642] Development of Sensor for Detection of Microplastics

Category: Hardware	**Theme:** Miscellaneous
Organization: University School of Automation & Robotics (USAR), GGSIPU	**Domain:** Environmental Monitoring

Problem Statement:

Develop portable low-cost sensors for detecting microplastics in water sources. Existing laboratory methods are expensive and unsuitable for real-time field monitoring.

Expected Solution:

Optical sensor-based detection system using light scattering, fluorescence, or absorbance combined with ML-based particle identification.

Technologies & Dataset:

Tech: Optical Sensors, Machine Learning, Embedded Systems, IoT, Edge AI | **Dataset:** Standard microplastic samples, water quality datasets

Deliverables:

Portable detection prototype, mobile/cloud monitoring interface, validation report

43. [GGSIPU2643] Development of Low-Cost Camera-Based Automated Beach Sand Grain Size Mapping System

Category: Hardware	**Theme:** Miscellaneous
Organization: University School of Automation & Robotics (USAR), GGSIPU	**Domain:** Coastal Monitoring

Problem Statement:

Manual sediment sampling for beach grain analysis is expensive and time-consuming. A camera-based automated system is required for continuous monitoring.

Expected Solution:

Camera and GNSS-enabled system with image processing algorithms to estimate grain size distribution and classify beach categories.

Technologies & Dataset:

Tech: Computer Vision, Image Processing, GPS/GNSS, Machine Learning, Embedded Systems | **Dataset:** Beach imagery datasets, sediment samples

Deliverables:

Camera prototype, AI estimation model, GPS mapping system, validation report

44. [GGSIPU2644] Embedded Intelligent Microscopy System for Identification and Counting of Microscopic Marine Organisms

Category: Hardware	**Theme:** Smart Automation
Organization: University School of Automation & Robotics (USAR), GGSIPU	**Domain:** Marine Biology

Problem Statement:

Manual microscopic identification of marine organisms is labor intensive and error prone. Need embedded AI-based microscopy solution for biodiversity assessment.

Expected Solution:

Embedded microscopy system using lightweight deep learning models for species identification and automated counting.

Technologies & Dataset:

Tech: Embedded AI, Deep Learning, Computer Vision, Microscopy, Edge Computing | **Dataset:** Marine microscopy image datasets

Deliverables:

AI microscopy prototype, identification model, counting module, reporting system

45. [GGSIPU2645] Smart Waste Segregation and Recycling System

Category: Hardware	**Theme:** Clean & Green Technology
Organization: University School of Automation & Robotics (USAR), GGSIPU	**Domain:** Waste Management

Problem Statement:

Improper household waste segregation reduces recycling efficiency and increases landfill burden. A smart system is required to classify waste at source and integrate with municipal waste management systems.

Expected Solution:

IoT-enabled waste segregation device using sensors and ML models to automatically classify organic, recyclable, and hazardous waste. Mobile application for monitoring disposal behaviour and incentives.

Technologies & Dataset:

Tech: IoT, Computer Vision, CNN Models, Sensors, Embedded Systems, Mobile Apps, Cloud Computing | **Dataset:** Household waste image datasets, municipal waste records

Deliverables:

Working prototype, mobile application, waste monitoring dashboard, classification model

46. [GGSIPU2646] Disaster Response Drone for Remote Areas

Category: Hardware	**Theme:** Robotics and Drones
Organization: University School of Automation & Robotics (USAR), GGSIPU	**Domain:** Disaster Management

Problem Statement:

Develop an autonomous drone system capable of delivering medical supplies and communication devices during disasters such as floods and earthquakes in inaccessible regions.

Expected Solution:

AI-enabled autonomous drone with navigation, obstacle avoidance, payload carrying capability (minimum 5 kg), and disaster management coordination application.

Technologies & Dataset:

Tech: UAV Technology, AI Navigation, Computer Vision, GPS, IoT, Embedded Systems, Ardupilot | **Dataset:** Satellite maps, disaster simulation datasets, terrain data

Deliverables:

Functional drone prototype, control application, navigation algorithm, field testing report

47. [GGSIPU2647] Renewable Energy Monitoring System for Microgrids

Category: Hardware	**Theme:** Renewable / Sustainable Energy
Organization: University School of Automation & Robotics (USAR), GGSIPU	**Domain:** Energy Management

Problem Statement:

Rural solar and wind microgrids require efficient monitoring of generation, storage, and consumption patterns.

Expected Solution:

IoT-based monitoring system providing real-time energy analytics, fault detection, and maintenance alerts.

Technologies & Dataset:

Tech: IoT, Sensors, Renewable Energy Monitoring, Cloud Analytics, Embedded Systems | **Dataset:** Energy generation datasets, microgrid operational data

Deliverables:

Hardware prototype, monitoring dashboard, mobile application

48. [GGSIPU2648] Improved Onion Storage Technology for Enhancing Shelf Life of Onions

Category: Hardware	**Theme:** Agriculture, FoodTech & Rural Development
Organization: University School of Automation & Robotics (USAR), GGSIPU	**Domain:** Post-Harvest Technology

Problem Statement:

Conventional onion storage results in 30–40% losses due to temperature, humidity variations, sprouting, and rotting.

Expected Solution:

Low-cost improved storage infrastructure using innovative materials, ventilation, solar/plasma/IoT technologies for maintaining optimum storage conditions.

Technologies & Dataset:

Tech: IoT Sensors, Climate Control, Agriculture Technology, Solar Systems, AI Monitoring | **Dataset:** Temperature-humidity datasets, onion storage data

Deliverables:

Storage prototype, monitoring system, shelf-life analysis, deployment plan

49. [GGSIPU2649] Automated High-Current Short-Circuit Test System for MCB Compliance with IEC 60898-1:2015

Category: Hardware	**Theme:** Smart Automation
Organization: University School of Automation & Robotics (USAR), GGSIPU	**Domain:** Electrical Testing

Problem Statement:

Existing manual testing of Miniature Circuit Breakers introduces safety risks and inconsistency during high current fault testing up to 10,000A.

Expected Solution:

Automated testing machine controlling current, voltage, impedance, waveform acquisition, and generating compliance reports according to IEC standards.

Technologies & Dataset:

Tech: PLC, Industrial Automation, High Current Electronics, DAQ Systems, HMI, Embedded Control | **Dataset:** Electrical testing parameters, MCB test records

Deliverables:

Automated test station, control software, safety system, compliance report generator

50. [GGSIPU2650] Automated Specimen Preparation System for Testing Cable Samples as per IS 10810 and IS 7098

Category: Hardware	**Theme:** Miscellaneous
Organization: University School of Automation & Robotics (USAR), GGSIPU	**Domain:** Industrial Automation

Problem Statement:

Manual cable specimen preparation introduces errors and inconsistency during electrical cable testing.

Expected Solution:

Automated cable feeding, cutting, stripping, shaping, and specimen preparation machine with PLC control.

Technologies & Dataset:

Tech: Industrial Automation, Robotics, PLC, Machine Vision, Sensors | **Dataset:** Cable sample specifications, IS standard requirements

Deliverables:

Prototype machine, control system, technical documentation

51. [GGSIPU2651] Non-Destructive Testing Method for Gold Jewellery and Artefacts

Category: Hardware	**Theme:** Miscellaneous
Organization: University School of Automation & Robotics (USAR), GGSIPU	**Domain:** Material Testing & Metrology

Problem Statement:

Fire assay testing of gold is accurate but destructive. Existing non-destructive methods struggle with heterogeneous jewellery compositions.

Expected Solution:

Develop accurate non-destructive gold purity assessment method comparable to fire assay while eliminating sample damage and harmful emissions.

Technologies & Dataset:

Tech: Spectroscopy, XRF, AI/ML, Material Science, Signal Processing | **Dataset:** Gold composition datasets, spectroscopy datasets

Deliverables:

Testing methodology, prototype system, validation report, scientific documentation

52. [GGSIPU2652] Detection and Prevention of Tampering in Weighing and Measuring Instruments

Category: Hardware	**Theme:** Miscellaneous
Organization: University School of Automation & Robotics (USAR), GGSIPU	**Domain:** IoT Security / Metrology

Problem Statement:

Digital weighing instruments and meters can be manipulated through hardware modifications, firmware changes, or unauthorized access.

Expected Solution:

Embedded tamper detection system with secure logging, firmware verification, remote monitoring, and regulatory alerts.

Technologies & Dataset:

Tech: Embedded Systems, IoT Security, Blockchain, Secure Boot, Sensors, Digital Forensics | **Dataset:** Device logs, tampering scenarios, firmware integrity datasets

Deliverables:

Tamper detection prototype, monitoring dashboard, validation report

53. [GGSIPU2653] Smart Agriculture for Efficient Cultivation in Hilly Regions

Category: Hardware	**Theme:** Agriculture, FoodTech & Rural Development
Organization: University School of Automation & Robotics (USAR), GGSIPU	**Domain:** Smart Irrigation

Problem Statement:

Farmers in hilly regions face water scarcity and inefficient irrigation due to climate variability.

Expected Solution:

Sensor-based irrigation system integrating rainwater harvesting, crop intelligence, and automated water control.

Technologies & Dataset:

Tech: IoT, Soil Sensors, ESP32/Arduino, Cloud Dashboard, Agriculture AI | **Dataset:** Soil moisture data, crop water requirement datasets

Deliverables:

Smart irrigation prototype, farmer dashboard, rainwater monitoring system

54. [GGSIPU2654] Cost Effective Solution for Detecting Breakage of Low Voltage AC Distribution Overhead Conductors

Category: Hardware	**Theme:** Disaster Management
Organization: University School of Automation & Robotics (USAR), GGSIPU	**Domain:** Electrical Safety

Problem Statement:

Broken live overhead conductors create electrocution hazards because conventional protection systems fail to detect low fault currents.

Expected Solution:

IoT and AI-based edge intelligence system to detect conductor breakage and immediately isolate supply with remote alerts.

Technologies & Dataset:

Tech: IoT, Edge AI, Current Sensors, Smart Grid Technology, Embedded Systems | **Dataset:** Electrical fault datasets, distribution network data

Deliverables:

Detection hardware, alert system, control dashboard

55. [GGSIPU2655] Improving Renewable Energy Hosting Capacity in Distribution Feeders

Category: Hardware	**Theme:** Renewable / Sustainable Energy
Organization: University School of Automation & Robotics (USAR), GGSIPU	**Domain:** Smart Grid

Problem Statement:

High rooftop solar penetration causes phase imbalance and poor voltage profiles in distribution networks.

Expected Solution:

IoT-enabled edge controller for dynamic phase balancing based on renewable generation and demand conditions.

Technologies & Dataset:

Tech: Smart Grid, IoT, Power Electronics, Edge Computing, Load Balancing Algorithms | **Dataset:** Grid voltage data, solar generation data

Deliverables:

Controller prototype, monitoring software, performance analysis

56. [GGSIPU2656] Design and Implementation of Solar-Powered Dewatering in Mining Operations

Category: Hardware	**Theme:** Renewable / Sustainable Energy
Organization: University School of Automation & Robotics (USAR), GGSIPU	**Domain:** Mining Technology

Problem Statement:

Existing mining dewatering systems rely on electricity and diesel generators leading to high operational costs and emissions.

Expected Solution:

Solar-powered pumping system under OPEX model with hybrid backup and vendor-managed operation.

Technologies & Dataset:

Tech: Solar PV, Pump Systems, IoT Monitoring, Energy Management Systems | **Dataset:** Mine water flow data, energy consumption data

Deliverables:

Solar dewatering prototype, operational model, cost analysis

57. [GGSIPU2657] Detection and Prevention of Unauthorized Use of Electric Fences

Category: Hardware	**Theme:** Smart Automation
Organization: University School of Automation & Robotics (USAR), GGSIPU	**Domain:** Electrical Safety

Problem Statement:

Unauthorized electric fences pose safety risks and may cause accidental injuries or fatalities.

Expected Solution:

Smart hardware system to detect illegal electric fence usage and provide alerts.

Technologies & Dataset:

Tech: IoT, Voltage Sensors, Embedded Systems, Communication Modules | **Dataset:** Electrical fence monitoring data

Deliverables:

Detection device, alert mechanism, monitoring dashboard

58. [GGSIPU2658] Earthquake Stabilised Dialysis System for Patient Safety During Seismic Events

Category: Hardware	**Theme:** Disaster Management
Organization: University School of Automation & Robotics (USAR), GGSIPU	**Domain:** Medical Devices

Problem Statement:

Dialysis infrastructure lacks earthquake-resistant mechanisms, risking treatment interruption during seismic events.

Expected Solution:

Stabilization mechanism enabling uninterrupted dialysis operation during earthquakes.

Technologies & Dataset:

Tech: Mechanical Engineering, Medical Devices, Sensors, Vibration Monitoring | **Dataset:** Earthquake vibration datasets

Deliverables:

Stabilized dialysis prototype, safety validation report

59. [GGSIPU2659] Solution for Non-Revenue Water Loss and Water Conservation

Category: Hardware	**Theme:** Miscellaneous
Organization: University School of Automation & Robotics (USAR), GGSIPU	**Domain:** Water Management

Problem Statement:

Water supply systems suffer losses due to leakage, inefficient usage, and poor conservation practices.

Expected Solution:

Smart water monitoring system detecting losses, improving awareness, and enabling wastewater reuse.

Technologies & Dataset:

Tech: IoT Water Sensors, Smart Metering, Data Analytics, Mobile Apps | **Dataset:** Water consumption datasets, leakage records

Deliverables:

Monitoring prototype, analytics dashboard, conservation framework

60. [GGSIPU2660] Grey Water Management and Reuse Wetland Management System

Category: Hardware	**Theme:** Clean & Green Technology
Organization: University School of Automation & Robotics (USAR), GGSIPU	**Domain:** Water Conservation

Problem Statement:

Improper greywater disposal affects environmental sustainability. Need solutions for treatment, reuse, and wetland management.

Expected Solution:

Smart greywater treatment and reuse system integrated with monitoring sensors.

Technologies & Dataset:

Tech: IoT Sensors, Water Treatment Technology, Automation, Environmental Monitoring | **Dataset:** Water quality datasets

Deliverables:

Treatment prototype, monitoring platform, reuse strategy

61. [GGSIPU2661] AI-Powered Skill Gap Intelligence & Curriculum Alignment Engine Linking Live Industry Demand to University Syllabi at USAR, GGSIPU

Category: Software	**Theme:** Industry-Academia Convergence, Employability & NEP 2020 Implementation
Organization: University School of Automation & Robotics (USAR), GGSIPU East Delhi Campus	**Domain:** Natural Language Processing (NLP), Large Language Models, Labour Market Analytics, Recommendation Systems, Knowledge Graphs, Educational Data Mining

Problem Statement:

Engineering curricula are revised on a multi-year cycle by a Board of Studies, while industry hiring requirements change every few months, so by the time a syllabus unit reaches a classroom the tool, framework or practice it teaches may already be obsolete. At USAR, GGSIPU East Delhi Campus, four B.Tech programmes in automation, robotics, AI and allied areas feed into a Delhi-NCR job market whose actual demand signal is scattered across job portals, company job descriptions, internship postings, freelance briefs and campus placement criteria that nobody systematically reads back into course design. The result is a measurable gap: students clear semesters yet fail first-round screening on skills the syllabus never named, and faculty revising a syllabus have opinion rather than evidence. Develop an AI-powered Skill Gap Intelligence & Curriculum Alignment Engine that continuously ingests live job market signals, extracts and normalises the skills actually being demanded for entry-level roles relevant to each programme, maps them against declared Course Outcomes and Programme Outcomes, and produces evidence-backed recommendations for curriculum revision, elective design, value-added courses and per-student learning paths.

Expected Solution:

Develop a platform that scrapes or ingests job descriptions, internship postings and placement criteria through permitted APIs and public sources; uses NLP and LLMs to extract, disambiguate and normalise skills into a canonical skill taxonomy mapped to NSQF and NEP 2020 credit structures; builds a knowledge graph linking skills to job roles, courses, course outcomes, laboratory experiments and assessment items; and quantifies gap severity by combining demand frequency, salary signal, growth trend and current syllabus coverage. The system should generate a Board of Studies dashboard that flags stale, redundant and missing units with citations to the underlying demand evidence; recommend new electives, laboratory experiments and value-added modules with draft outcomes; and produce a per-student gap profile from academic records and self-declared projects, with a personalized learning path drawn from open courseware and a projected employability delta. The platform must show its evidence for every recommendation and must never reduce a student to a single opaque employability score.

Technologies & Dataset:

Tech: Python, FastAPI, Hugging Face Transformers, Gemini/OpenAI/Llama, LangChain, spaCy, Sentence Transformers, Neo4j Knowledge Graph, FAISS/ChromaDB, Scikit-learn, Pandas, Streamlit/React, PostgreSQL, Elasticsearch, Google Apps Script (faculty dashboard), Docker, GitHub | **Dataset:** Public job posting APIs and portals (where permitted by terms of use), NCS and Skill India datasets, ESCO and O*NET occupational taxonomies, NSQF Qualification Packs, AICTE Model Curriculum, USAR/GGSIPU syllabi and Course Outcome documents, historical campus placement criteria and offer data (anonymised), Open Courseware catalogues

Deliverables:

Skill extraction and normalisation pipeline; Canonical skill taxonomy mapped to NSQF; Skill-Course-Outcome knowledge graph; Board of Studies curriculum gap dashboard with evidence citations; Draft elective and value-added course recommendations; Per-student gap profile and personalized learning path module; Trend and forecasting reports for the Training & Placement Cell; Technical documentation; Source code repository; Deployment guide; Demo video

62. [GGSIPU2662] Verified Skill Passport and Industry Micro-Project Marketplace Connecting Delhi-NCR MSMEs with USAR, GGSIPU Student Teams

Category: Software	**Theme:** Industry-Academia Convergence, Experiential Learning & Employability
Organization: University School of Automation & Robotics (USAR), GGSIPU East Delhi Campus	**Domain:** Web Platforms, Recommendation Systems, Cryptographic Verification, Workflow Automation, Learning Analytics, NEP 2020 Apprenticeship-Embedded Degrees

Problem Statement:

Two shortages sit next to each other and never meet. Students at USAR, GGSIPU East Delhi Campus need authentic industry work to be employable, but mostly accumulate certificates that prove attendance rather than capability, and internship access is skewed toward those with existing networks. Meanwhile thousands of MSMEs, startups and NGOs across Delhi-NCR have small, real, unglamorous technical problems - a dashboard, a data cleanup, an automation script, a computer vision proof of concept - that they cannot afford to outsource and cannot scope well enough to hire for. NEP 2020 and apprenticeship-embedded degree structures now formally permit such work to carry academic credit, but there is no trusted infrastructure to scope it, supervise it, verify it or convert it into credit. Develop a platform that lets a verified organisation post a scoped micro-project, matches it to a student team and a faculty mentor, tracks contribution through actual work artefacts rather than self-reporting, and issues a tamper-evident Verified Skill Passport that a recruiter can check in seconds and that maps cleanly onto the university's credit and internship requirements.

Expected Solution:

Develop a web platform supporting organisation verification, AI-assisted scoping that converts a vague industry request into a bounded deliverable with acceptance criteria and a realistic effort estimate, skill-based matching of student teams to projects, faculty mentor assignment, milestone and payment or stipend tracking, and structured client feedback. Contribution evidence should be captured from actual artefacts - repository commits, design files, documents, deployment links, demo recordings - rather than declarations, and each verified competency should be issued as a cryptographically hashed, QR-verifiable credential in the manner of GGSIPU2609, with public verification and revocation. The platform must include a credit-mapping module that translates completed project hours and outcomes into internship or project credits under NEP 2020 and university regulations for faculty approval, a dispute and grievance workflow, and an equity mechanism that deliberately surfaces first-generation learners and students without prior industry exposure instead of ranking purely on past performance.

Technologies & Dataset:

Tech: React/Next.js, Node.js or FastAPI, PostgreSQL, Redis, SHA-256 hashing and optional Polygon/Hyperledger anchoring, GitHub/GitLab APIs, Gemini/OpenAI/Llama for scoping assistance, Sentence Transformers for matching, Google Workspace APIs, QR Code libraries, Docker, GitHub | **Dataset:** MSME and startup registries, synthetic industry project briefs, USAR student academic and project records (with consent), skill taxonomies from GGSIPU2627, historical internship and industrial training records, NEP 2020 credit framework documents, synthetic test data

Deliverables:

Micro-project marketplace platform; Organisation verification module; AI-assisted project scoping assistant; Team and mentor matching engine; Artefact-based contribution tracking; Verified Skill Passport with QR and hash verification; Public recruiter verification portal; NEP credit mapping and faculty approval workflow; Equity and access dashboard; Technical documentation; Source code repository; Deployment guide; Demo video

63. [GGSIPU2663] Agentic AI Placement Readiness, Simulated Hiring Pipeline and Fair Assessment System for the USAR, GGSIPU Training & Placement Cell

Category: Software	**Theme:** Industry-Academia Convergence, Employability & Responsible AI
Organization: University School of Automation & Robotics (USAR), GGSIPU East Delhi Campus	**Domain:** Agentic AI, Large Language Models, Speech AI, Conversational AI, Assessment Analytics, Explainable AI, Algorithmic Fairness

Problem Statement:

Campus placement preparation at USAR, GGSIPU East Delhi Campus is compressed into a few weeks, delivered to hundreds of students at once by a small Training & Placement Cell, and reduced to generic aptitude drills, while the actual failure points are individual and diagnosable: a resume that no applicant tracking system parses, weak articulation of one's own project, a specific data structures blind spot, freezing in a technical viva, or code-mixed communication anxiety that has nothing to do with technical ability. Students who most need practice get the least, because mock interviews are limited by faculty and alumni availability. Develop an agentic AI system that runs an end-to-end simulated hiring pipeline for a target role - resume screening, aptitude and technical assessment, coding round, technical interview, HR interview - with specialised agents for each stage, and returns specific, evidence-linked, actionable feedback rather than a score. The system must be explicitly designed to reduce rather than reproduce hiring bias, and must be positioned as practice, never as a gatekeeping mechanism for real placement opportunities.

Expected Solution:

Develop a multi-agent platform where a Job Description Agent constructs a realistic role profile from live postings; a Resume Agent performs ATS-style parsing and returns concrete rewrite suggestions tied to evidence in the student's own record; an Assessment Agent generates role-calibrated aptitude, domain and coding problems with adaptive difficulty; an Interviewer Agent conducts a voice or text technical interview that probes the student's actual submitted projects and follows up on shallow answers; an HR Agent evaluates situational responses; and an Evaluator Agent synthesises a diagnostic report mapping each weakness to a specific remediation resource and a practice schedule. The platform should support code-mixed Hindi-English interaction, provide the T&P Cell with an aggregate cohort readiness dashboard identifying which skills need a batch-level intervention, and include a mandatory fairness module that audits scoring across gender, medium of instruction and accent, publishes disparity metrics, and blocks deployment of any scoring component that fails a stated disparity threshold. All feedback must be transparent about the model's uncertainty, and no student-level score may be exposed to recruiters or used for shortlisting.

Technologies & Dataset:

Tech: Python, FastAPI, LangGraph, CrewAI, Gemini/OpenAI/Llama, Whisper, TTS/STT APIs, Hugging Face Transformers, Sentence Transformers, code execution sandboxes, PostgreSQL, Vector Databases, React, Fairlearn or AIF360, Docker, GitHub | **Dataset:** Public job descriptions, open-source coding problem sets, publicly available interview question banks, anonymised USAR placement and assessment history (with consent), student resumes and project records (with consent), synthetic interview transcripts, speech datasets for Indian-accented and code-mixed English

Deliverables:

Multi-agent simulated hiring pipeline; ATS-style resume analyser with rewrite suggestions; Adaptive assessment and coding round engine with sandboxed execution; Voice and text technical interview agent; Diagnostic feedback report generator with remediation paths; T&P Cell cohort readiness dashboard; Fairness and bias audit module with published disparity metrics; Responsible use policy; Technical documentation; Source code repository; Deployment guide; Demo video

64. [GGSIPU2664] Agentic AI Security Operations Centre for Autonomous Threat Detection, Triage and Guarded Response across University Networks (USAR, GGSIPU)

Category: Software	**Theme:** Cybersecurity, Agentic AI & Critical Infrastructure Defence
Organization: University School of Automation & Robotics (USAR), GGSIPU East Delhi Campus	**Domain:** Agentic AI, Cybersecurity, Security Operations, Anomaly Detection, Large Language Models, Explainable AI, Incident Response

Problem Statement:

Universities are soft, high-value targets: GGSIPU East Delhi Campus and USAR run an ERP holding student personal and financial data, examination and result systems, open campus Wi-Fi with thousands of unmanaged personal devices, laboratory machines and research servers, CCTV and IoT controllers, and a Google Workspace tenant - defended by a small IT cell with no round-the-clock security staffing. Attacks on Indian educational and research institutions, including ransomware and data exfiltration, have risen sharply, and CERT-In directions require reporting a cyber incident within six hours of noticing it, a deadline no manual process reliably meets at 2 a.m. Meanwhile a commercial managed SOC is priced beyond most university budgets. Develop an agentic AI Security Operations Centre that autonomously monitors logs and telemetry, detects and correlates suspicious activity, triages alerts with reasoning a human analyst can audit, proposes and where authorised executes bounded containment actions, and drafts the compliance-ready incident report - so that a two-person IT cell effectively operates a twenty-four-hour SOC. The system is defensive: it must not include capability for developing, weaponising or launching attacks against third-party systems.

Expected Solution:

Develop a multi-agent defensive platform in which specialised agents divide the work: a Collection Agent normalises telemetry from firewalls, endpoints, authentication logs, DNS, DHCP, ERP application logs, mail gateways and Workspace audit logs; a Detection Agent combines signature rules with unsupervised anomaly detection and user and entity behaviour analytics to surface deviations such as impossible-travel logins, credential-stuffing patterns, lateral movement, unusual data egress volumes and off-hours privileged access; a Correlation Agent assembles related alerts into a single incident narrative mapped to MITRE ATT&CK tactics; a Triage Agent assigns severity using asset criticality and blast radius, suppresses known-benign patterns and writes an evidence-linked explanation; a Response Agent proposes containment from a pre-approved, scoped playbook library - isolate host, disable account, block indicator, force credential reset - with strict human-in-the-loop authorisation for any action above a defined blast-radius threshold, full rollback, and a hard prohibition on actions reaching outside university-owned assets; and a Reporting Agent drafts the CERT-In-format incident report and post-incident timeline. The platform must include a deliberate anti-abuse design note explaining how the agent is prevented from being repurposed offensively, and how prompt injection embedded in ingested log data is neutralised so that a malicious log line cannot steer the agent.

Technologies & Dataset:

Tech: Python, FastAPI, LangGraph, CrewAI, Gemini/OpenAI/Llama, Wazuh or Elastic Security, Suricata, Zeek, Sigma rules, MITRE ATT&CK and Atomic Red Team mappings, Scikit-learn, PyTorch, Isolation Forest and autoencoder models, Kafka, Elasticsearch, PostgreSQL, Grafana, TheHive or Shuffle SOAR, Docker, Kubernetes, GitHub | **Dataset:** Public security datasets including CIC-IDS2017, UNSW-NB15, LANL authentication logs and CERT insider threat data; MITRE ATT&CK knowledge base; CERT-In advisories and reporting formats; synthetic university network telemetry; anonymised campus network logs generated in an isolated laboratory environment with institutional permission

Deliverables:

Agentic SOC platform; Telemetry collection and normalisation pipeline; Hybrid detection engine with UEBA; Incident correlation and ATT&CK mapping module; Explainable triage engine with evidence trails; Guarded response playbook library with human-in-the-loop authorisation and rollback; CERT-In compliant incident report generator; Analyst dashboard; Prompt-injection and anti-abuse hardening note; Technical documentation; Source code repository; Deployment guide for a small university IT cell; Demo video

65. [GGSIPU2665] Agentic AI Platform for Continuous Attack Surface Discovery, Risk Prioritisation and Preemptive Hardening of Institutional Digital Assets (USAR, GGSIPU)

Category: Software	**Theme:** Cybersecurity, Agentic AI & Preventive Risk Management
Organization: University School of Automation & Robotics (USAR), GGSIPU East Delhi Campus	**Domain:** Agentic AI, Attack Surface Management, Vulnerability Intelligence, Software Supply Chain Security, DevSecOps, Risk Analytics

Problem Statement:

Most institutional breaches begin not with a sophisticated exploit but with something nobody knew was exposed: a departmental portal spun up for one admission cycle and never decommissioned, an old subdomain still resolving to a dead host, a Google Apps Script web app deployed with anyone-with-the-link access, a laboratory server with default credentials, a student project repository with a live API key committed, an unpatched library three versions behind. USAR and GGSIPU East Delhi Campus accumulate exactly this kind of debt because assets are created by many hands - departments, societies, individual faculty, student teams - and inventoried by none. Traditional vulnerability scanners then invert the problem by producing thousands of undifferentiated findings that a two-person IT cell cannot act on. Develop an agentic AI platform that continuously discovers what the institution actually exposes, reasons about which exposures genuinely matter given real-world exploitation activity and institutional context, and produces a small, ordered, executable hardening plan. Scope is strictly limited to assets the institution owns or is explicitly authorised to assess; the platform must technically enforce that boundary.

Expected Solution:

Develop a platform where a Discovery Agent continuously enumerates institution-owned domains, subdomains, exposed services, cloud and Workspace assets, publicly readable Drive and Apps Script deployments, and shadow IT, maintaining a live asset inventory with ownership attribution; an SBOM Agent generates software bills of materials for institutional applications and repositories and maps dependencies to known vulnerability advisories; a Prioritisation Agent ranks findings by combining severity with real-world exploitation signals such as CISA KEV and EPSS, asset criticality, data sensitivity, internet reachability and existing compensating controls, so that a handful of genuinely urgent items rise above the noise; a Remediation Agent drafts specific, testable fixes - configuration changes, version bumps, access-scope corrections, draft pull requests - with an estimated effort and a rollback plan; and a Validation Agent confirms remediation using non-intrusive verification only, or by replaying published detection signatures inside an isolated laboratory range that mirrors the asset. The platform must enforce an authorisation boundary through a signed scope file so that no discovery or verification activity can be directed at systems outside the approved institutional inventory, must rate-limit and log every external interaction, and must produce a management-facing risk trend report suitable for presentation to university authorities.

Technologies & Dataset:

Tech: Python, FastAPI, LangGraph, CrewAI, Gemini/OpenAI/Llama, OWASP Dependency-Track, Trivy, Syft and Grype, Nuclei with authorised scope enforcement, Amass or Subfinder, CycloneDX and SPDX SBOM standards, NVD, CISA KEV and EPSS feeds, OSV database, PostgreSQL, Elasticsearch, Grafana, GitHub Actions, Docker, Kubernetes | **Dataset:** National Vulnerability Database, OSV, CISA Known Exploited Vulnerabilities catalogue, EPSS scores, CERT-In advisories, institution-owned asset inventory and DNS records (with written authorisation), USAR and GGSIPU application repositories and Apps Script deployments (with permission), synthetic enterprise asset datasets

Deliverables:

Continuous attack surface discovery engine with ownership attribution; Live institutional asset inventory; SBOM generation and dependency risk module; Context-aware exploitability prioritisation engine; Draft remediation plans with rollback guidance; Non-intrusive validation module and isolated laboratory range configuration; Signed authorisation scope enforcement mechanism; Executive risk trend dashboard; Technical documentation; Source code repository; Deployment guide; Demo video

66. [GGSIPU2666] Agentic AI Defence Against Phishing, Deepfake Voice and Social Engineering Attacks Targeting University Students, Faculty and Administration (USAR, GGSIPU)

Category: Software	**Theme:** Cybersecurity, Agentic AI, Synthetic Media Forensics & Human-Centred Security
Organization: University School of Automation & Robotics (USAR), GGSIPU East Delhi Campus	**Domain:** Agentic AI, Multimodal AI, Deepfake and Synthetic Media Detection, Natural Language Processing, Speech Forensics, Human-Centred Security, Security Awareness

Problem Statement:

The most effective attacks on a university bypass its network entirely and target its people. Students at USAR, GGSIPU East Delhi Campus receive fraudulent fee-payment links timed to the actual fee deadline, fake internship and placement offers demanding a registration fee, scholarship-disbursal scams and account-verification lures on WhatsApp and Telegram; administrative staff receive spoofed instructions purporting to come from a Dean or the Registrar; and generative AI has now made voice cloning cheap enough that a finance section can receive a convincing phone call in a senior official's voice authorising an urgent transfer, or a student can receive a video call from a cloned relative. Existing spam filters miss these because they arrive over consumer messaging channels, are written in code-mixed Hindi-English, and are individually crafted rather than mass-mailed. Develop an agentic AI defence platform that detects social engineering across the channels a university community actually uses, verifies whether audio or video purporting to be an institutional official is synthetic, warns the target in the moment with a plain-language explanation, and routes confirmed incidents into an institutional response and awareness loop.

Expected Solution:

Develop a multimodal, multi-agent platform in which an Ingestion Agent accepts user-forwarded messages, emails, links, screenshots, audio and video across email, WhatsApp, SMS and web submission; a Text Agent analyses linguistic and structural signals of social engineering - urgency, authority impersonation, payment redirection, credential harvesting, look-alike domains - with strong performance on code-mixed Hindi-English; a Link and Infrastructure Agent inspects URLs, redirect chains, domain age, certificate anomalies and look-alike registrations in a sandbox without executing payloads; a Media Forensics Agent screens audio for voice cloning and video for face-swap and lip-sync manipulation, returning a calibrated confidence with visual or spectral evidence rather than a bare verdict; a Verification Agent checks whether a claimed institutional sender, portal, notice or payment channel matches the university's authoritative registry of official domains, accounts and payment routes; and an Explanation Agent returns a plain-language, low-jargon warning naming the specific cues, in Hindi or English. The platform should include a one-tap student reporting flow, an institutional dashboard showing live campaigns targeting the community, an automatic advisory broadcast when a campaign crosses a threshold, and an adaptive awareness training module that teaches from the actual attacks currently circulating on campus. Detection must be honest about uncertainty, must never mislabel a legitimate university communication without a safe appeal path, and the platform must not generate synthetic media or persuasive lures for any purpose other than clearly labelled, contained defensive training with institutional approval.

Technologies & Dataset:

Tech: Python, FastAPI, LangGraph, CrewAI, Gemini/OpenAI/Llama, Hugging Face Transformers, IndicBERT or MuRIL for code-mixed text, Whisper, wav2vec 2.0 and audio anti-spoofing models, PyTorch, OpenCV, deepfake detection backbones such as CLIP-ViT and EfficientNet, URL sandboxing and headless browsing, VirusTotal and URLhaus APIs, PostgreSQL, Redis, Elasticsearch, React, Docker, GitHub | **Dataset:** Public phishing corpora including PhishTank, OpenPhish and Nazario; ASVspoof and In-the-Wild audio spoofing datasets; FaceForensics++, Celeb-DF and DFDC deepfake video datasets; multilingual and code-mixed Indian social engineering samples collected with consent; CERT-In and RBI fraud advisories; university registry of official domains, handles and payment channels; synthetic campaign data

Deliverables:

Multimodal social engineering detection platform; Code-mixed Hindi-English text analysis module; Sandboxed link and infrastructure analysis module; Audio voice-clone and video deepfake forensics module with calibrated confidence and visual evidence; Authoritative institutional identity and payment channel registry with verification service; Plain-language in-the-moment warning interface; One-tap student reporting flow; Institutional campaign dashboard and advisory broadcast system; Adaptive awareness training module; Responsible use and non-generation policy; Technical documentation; Source code repository; Deployment guide; Demo video